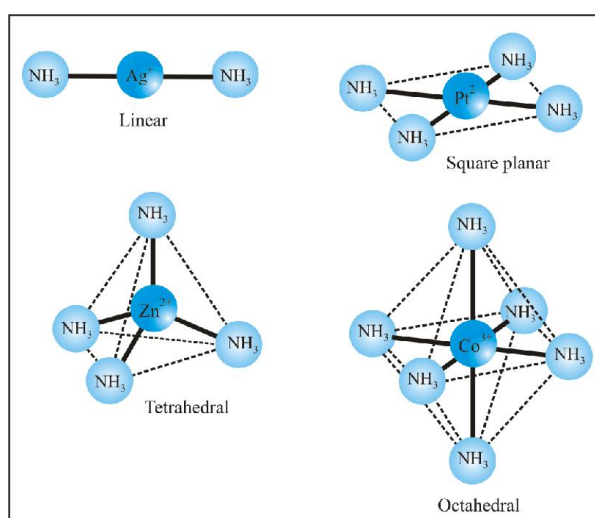


COORDINATION COMPOUNDS

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THEORY

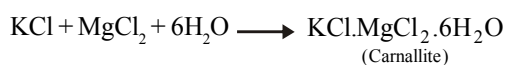
1. INTRODUCTION

Co-ordination compounds play a vital role. The importance can be realised that life would not have been possible without the existence of **chlorophyll (Mg - complex)** in plants and **haemoglobin (Fe- complex)** in the blood of human beings. The study of these compounds will enlarge our understanding of chemical bonding, physical properties such as magnetic properties of co-ordination compounds

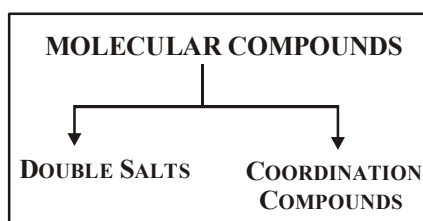
2. MOLECULAR OR ADDITION COMPOUNDS

When solution containing two or more simple stable compounds in molecular proportions are allowed to evaporate, crystals of new substances called molecular or addition compounds are obtained.

Example

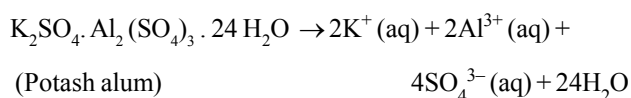
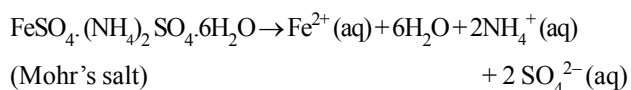


2.1 Types of Molecular compounds



2.1.1 Double Salt

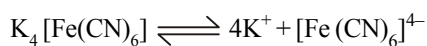
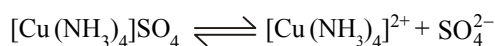
A double salt is a substance obtained by the combination of two different salts which crystallize together as a single substance but ionise as two distinct salts when dissolved in water. These salts lose their identity in solution i.e. when dissolved in water they give test of all the ions present in the salt. eg. Potash alum, Mohr's salt



2.2 Coordination Compounds

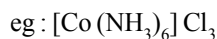
A coordination compound is a molecular compound that results from the combination of two or more simple molecular compounds and retains its identity in the solid as well as in dissolved state

Example

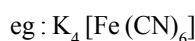


3. COORDINATION COMPOUNDS

A Co-ordination compound consists of a ligand, central atom, complex ion, a cation or an anion. The complex ion is generally written in a square box and the ion (cation or anion) is written outside complex ion.



[Complex ion] anion



cation [Complex ion]

General formula : $\text{A}_x[\text{ML}_n]/[\text{ML}_n]\text{B}_y$

where : M is the central metal atom/ion

L is the ligand

A is the cation

B is the anion

Some Important Terms pertaining to Coordination Compounds

3.1 Coordination entity

It is the central metal atom or ion which is bonded to a definite number of ions or molecules which is fixed. For example, in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, a coordination entity, six ammonia molecules are surrounded by three chloride ions.

3.2 Central atom/ion

It is the central cation that is surrounded and coordinately bonded to one or more neutral molecules or negatively charged ions in a definite geometrical arrangement. For example, in the complex $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, Co^{3+} represents the central metal ion which is positively charged and is coordinately bonded to six neutral NH_3 molecules within the coordination sphere. The central metal/ion is also referred to as Lewis acid.

3.3 Ligands

The ions or molecules bound to the central atom/ion in the coordination entity are called ligands. These may be simple ions such as Cl^- , small molecules such as H_2O or NH_3 , larger molecules such as $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$.

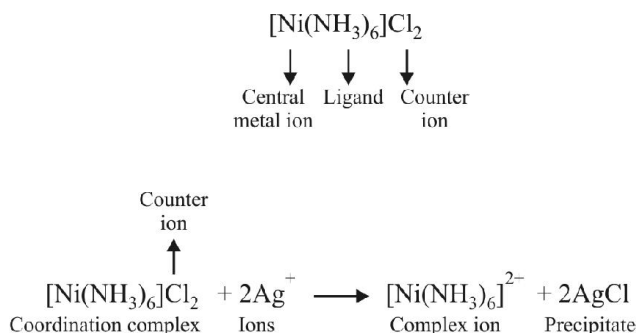
3.4 Co-ordination Number (C.N)

The number of atoms of the ligands that directly bound to the central metal atom or ion by co-ordinate bonds is known as the co-ordination number of the metal atom or ion. It is also equal to the secondary valency.

Complex	Co-ordination numbers
$\text{K}_4[\text{Fe}(\text{CN})_6]$	6
$[\text{Ag}(\text{CN})_2]^-$	2
$[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	4
$[\text{Ca}(\text{EDTA})]^{2-}$	6

3.5 Coordination sphere

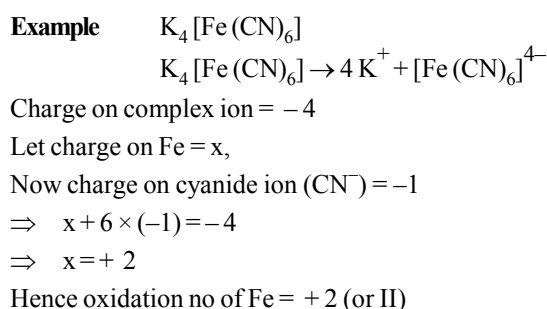
The central metal atom or ion and the ligands that are directly attached to it are enclosed in a square bracket. This had been called coordination sphere or first sphere of attraction. It behaves as a single unit because the ligands present in the coordination sphere are held tightly by the metal ion.

**3.6 Co-ordination Polyhedron**

A coordination polyhedron is the spatial arrangement of the ligand atoms that are directly attached to the central atom/ion. For example, $[\text{Co}(\text{NH}_3)_6]^{3+}$ is octahedral, $[\text{Ni}(\text{CO})_4]$ is tetrahedral and $[\text{PtCl}_4]^{2-}$ is square planar.

3.7 Oxidation Number of Central Metal Atom

It is defined as the charge that the central metal ion would carry if all the ligands are removed along with electron pairs. It is calculate as follows :

**3.8 Homoleptic and Heteroleptic Complexes**

Complexes in which central atom is coordinated with only one kind of ligands are called homoleptic complexes, eg. $[\text{Co}(\text{NH}_3)_6]^{3+}$. Complexes in which central atom is coordinated with more than one kind of ligands are called heteroleptic complexes, eg. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

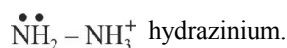
4. NOMENCLATURE OF COORDINATION COMPOUNDS**4.1 Nomenclature**

Following rules are adopted for naming a complex ion;

- Cations are named before anions
- Oxidation state (O.S.) of the central metal ion is denoted by Roman numeral.

Compound	Cation	O.S.	anion
CuCl	Copper	(I)	chloride
CuCl_2	Copper	(II)	chloride
FeCl_2	Iron	(II)	chloride
FeCl_3	Iron	(III)	chloride

- The names of ligands are given first followed by the name of the central metal ion.
- The names of ligands that are anions and ending with 'ide' are changed to 'o'
 'ite' are changed to 'ito'
 'ate' are changed to 'ato'
- Many ligands that are molecules carry the unmodified name
- Positive groups end in **-ium**



- (g) When there are several ligands of the same kind, we normally use the prefixes **di**, **tri**, **tetra**, **penta** and **hexa** to show the number of ligands of that type. An exception occurs when the name of the ligands includes a number, for example as in ethylenediamine (en). To avoid confusion in such cases, **bis**, **tris** and **tetrakis** are used instead of di, tri and tetra, and the name of the ligand is placed in brackets.

e.g. bis (ethylenediamine)

- (h) If anion is a complex, then metal ends with 'ate'

$[\text{Ni}(\text{CN})_4]^{2-}$: tetracyanonickelate (II) ion

lead	—	plumbate
gold	—	aurate
zinc	—	zincate
tin	—	stannate
silver	—	argentate
cobalt	—	cobaltate
iron	—	ferrate
aluminium	—	aluminate
manganese	—	manganate
copper	—	cuprate
chromium	—	chromate
platinum	—	platinate

- (i) If the complex contains two or more metal atoms, it is termed polynuclear. The bridging ligands which link the two metal atoms together are indicated by the prefix μ —

- (j) Ambidentate ligands may be attached through different atoms

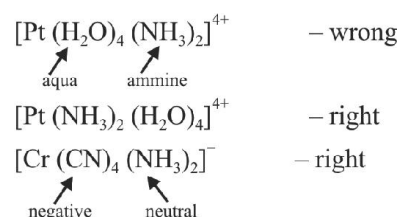
$\text{M} \leftarrow \text{NO}_2$ (NO_2 joined to metal M through N; it is **nitrito-N**)

$\text{M} \leftarrow \text{ONO}$ (NO_2 joined to metal M through O; it is **nitrito-O**)

Similarly, the SCN group may bond $\text{M} - \text{SCN}$ (thiocyanato) or $\text{M} - \text{NCS}$ (isothiocyanato).

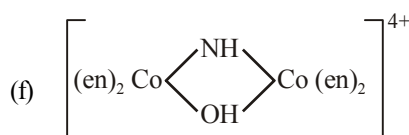
- (k) When writing (not naming) the formula of the complex :

- * Complex ion should be enclosed by square brackets and
- * Ligands are placed after metal in the alphabetical order but first negative ligands, then neutral, then positive.



EXAMPLE : Write the name of the following compounds ?

- $[\text{Fe}(\text{NH}_3)_6]\text{Cl}_3$
- $[\text{CoCl}(\text{NH}_3)_5]^{2+}$
- $[\text{CoSO}_4(\text{NH}_3)_4]\text{NO}_3$
- $[\text{Pt}(\text{Py})_4][\text{PtCl}_4]$
- $[\text{Cr}(\text{en})_3]\text{Cl}_3$



- Sol.** (a) Hexaammineiron (III) Chloride
 (b) Pentaamminechloridocobalt (III) ion
 (c) Tetraamminesulphatocobalt (III) nitrate
 (d) Tetrapyridineplatinum (II) tetrachloridoplatinate (II)
 (e) Tris (ethylenediamine) chromium (III) chloride
 (f) Tetrakis (ethylenediamine)- μ -hydroxo- μ -imidodicobalt (III) ion.

5. WERNER'S THEORY

Werner explained the nature of bonding in complexes and he concluded that in complexes the metal shows two different types of valency.

5.1 Primary Valency

The primary valency is equal to the oxidation state of the central metal atom or ion. These are non-directional.

EXAMPLE : What are the primary valency of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ & $[\text{Co}(\text{NH}_3)_6]^{3+}$?

Sol. $\text{K}_4[\text{Fe}(\text{CN})_6]$: Primary valency is 2.

$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$: Primary valency is 2.

5.2 Secondary Valency

The number of ligand atoms co-ordinated to the central metal atom is called secondary valency. These are directional and so a complex ion has a particular shape.

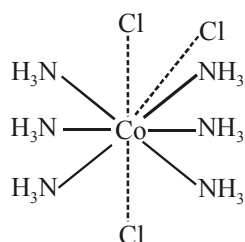
EXAMPLE : What are the secondary valency of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ & $\text{K}_4[\text{Fe}(\text{CN})_6]$?

Sol. In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ the secondary valency is 6.

$\text{K}_4[\text{Fe}(\text{CN})_6]$: six ligands are coordinated to Fe. Hence secondary valency is 6.

The primary valency is satisfied by ions attached to the complex ions. It is shown by dotted lines. Primary valency is also known as ionisable valency.

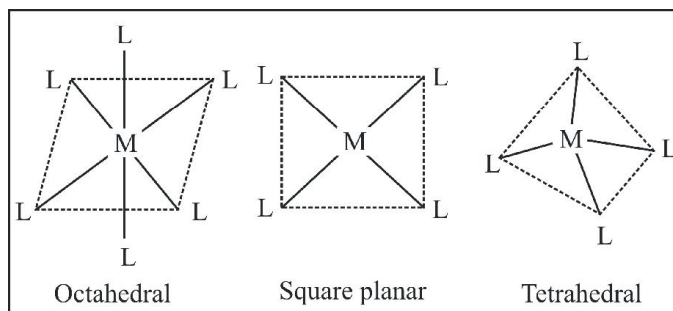
The secondary valency is satisfied by the ligands, they are non ionisable and are shown by a solid line $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ can be represented as



An anion present in co-ordination and ionization sphere is shown by

Every element tends to satisfy both its primary and secondary valencies. A negative ion when present in the coordination sphere shows a dual behaviour. It may satisfy both primary and secondary valencies.

be tetrahedral or square planar. This postulate predicted the existence of different types of isomerism in coordination compounds.



EXAMPLES:

Octahedral (C.N = 6)	Square planar (C.N = 4)	Tetrahedral (C.N. = 4)
$[\text{Cr}(\text{CH}_3)_6]^{3+}$	$[\text{Ni}(\text{CN})_4]^{2-}$	$[\text{Ni}(\text{CO})_4]$
$[\text{Co}(\text{NH}_3)_6]^{3+}$; $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	$[\text{Ni}(\text{NH}_3)_4]^{2+}$	$[\text{CuX}_4]^{2-}$; $[\text{ZnCl}_4]^{2-}$
$[\text{Fe}(\text{CN})_6]^{2-}$; $[\text{Fe}(\text{F}_6)]^{3-}$	$[\text{Cu}(\text{NH}_3)_4]^{2+}$	$[\text{NiX}_4]^{2-}$
$[\text{Pt}(\text{NH}_3)_6]^{4+}$; $[\text{PtCl}_6]^{2-}$		$\text{X} = \text{Cl}^-, \text{Br}^-, \text{I}^-$

Familiar C.N.'s of some common metal ions.

Univalent	C.N.	Divalent	C.N.
Ag^+	2	V^{2+}	6
Au^+	2, 4	Fe^{2+}	6
Ti^+	2	Co^{2+}	4, 6
Cu^+	2, 4	Ni^{2+}	4, 6
		Cu^{2+}	4, 6
		Zn^{2+}	4
		Pd^{2+}	4
		Pt^{2+}	4
		Ag^{2+}	4

Trivalent	C.N.	Tetravalent	C.N.
Sc^{3+}	6	Pt^{4+}	6
Cr^{3+}	6	Pd^{4+}	6
Fe^{3+}	6		
Co^{3+}	6		
Os^{3+}	6		
Ir^{3+}	6		
Au^{3+}	4		

The ligand which satisfy the secondary valencies are directed toward fixed positions in space. The geometry of the complex ion depends on the coordination number. If the metal has coordination number 6, the complex is octahedral, i.e. six positions around the metal are occupied by six donor atoms of the ligands octahedrally. On the other hand, if the coordination number is 4, the geometry of the complex may

6. EFFECTIVE ATOMIC NUMBER (EAN)

Sidwick proposed effective atomic number abbreviated as EAN, which is defined as the resultant number of electrons with the metal atom or ion after gaining electrons from the donor atoms of the ligands. The effective atomic number (EAN) generally coincides with the atomic number of next

inert gas in some cases. EAN is calculated by the following relation :

$$\text{EAN} = \text{Atomic number of the metal} - \text{number of electrons lost in ion formation} + \text{number of electrons gained from the donor atoms of the ligands. } (2 \times \text{CN})$$

The EAN values of various metals in their respective complexes are tabulated below :

Complex	Metal (Oxid. state)	At. No. of metal	Coordination number	Effective atomic number
$\text{K}_4[\text{Fe}(\text{CN})_6]$	+2	26	6	$(26 - 2) + (6 \times 2) = 36 [\text{Kr}]$
$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$	+2	29	4	$(29 - 2) + (4 \times 2) = 35$
$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	+3	27	6	$(27 - 3) + (6 \times 2) = 36 [\text{Kr}]$
$\text{Ni}(\text{CO})_4$	0	28	4	$(28 - 0) + (4 \times 2) = 36 [\text{Kr}]$
$\text{K}_2[\text{Ni}(\text{CN})_4]$	+2	28	4	$(28 - 2) + (4 \times 2) = 34.$

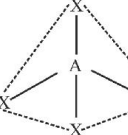
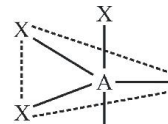
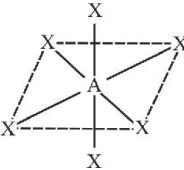
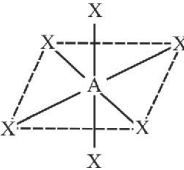
7. VALENCE BOND THEORY

The bonding in coordination compounds can be explained by Valence Bond Theory (VBT) since majority of the complexes formed by the transition metals have their d-orbitals incomplete. Valence bond takes into account the hybridisation of orbitals since penultimate d-orbitals are near in energy to s and p-orbitals of the outer most shell, various kinds of hybridization is possible.

VBT makes the following assumption

- A number of empty orbitals are available on the central metal ion which can accommodate electrons donated by the ligands. The number of empty d-orbitals is equal to the coordination number of the metal ion for the particular complex.
- The metal orbitals and ligand orbitals overlap to form strong bonds. Greater the extent of overlapping, more is the stability of the complex. Different orbitals (s, p or d) hybridize to give a set of equivalent hybridized orbital which take part in bonding with the ligands.
- Each ligand donates a pair of electrons to the central metal ion/atom.
- The non-bonding metal electrons present in the inner orbitals do not take part in chemical bonding.
- If the complex contains unpaired electrons, the complex is paramagnetic. If it does not contain unpaired electron, the complex is diamagnetic in nature.
- Under the influence of strong ligand (CN, CO) the electrons can be forced to pair up against the Hund's rule of multiplicity.

COMMON TYPES OF HYBRIDISATION

Coordination Number	Hybridization	Shape	Geometry
2	sp	Linear	$\text{X} - \text{A} - \text{X}$ 
4	sp^3	Tetrahedron	
4	dsp^2	Square planar	
5	sp^3d or dsp^3	Trigonal bipyramid	
6	d^2sp^3 or sp^3d^2	Octahedral	

NOTE:

In d^2sp^3 hybridisation, the inner d-orbitals (3d orbital) has been used for bonding, such complexes are called inner orbital complexes or low spin complexes.

In sp^3d^2 hybridisation the outer d-orbitals (4d orbital) has been used for bonding, such complex are called outer orbital complexes or high spin complexes.

The magnetic moment is given by $\sqrt{n(n+2)}$ BM where n is the number of unpaired electrons.

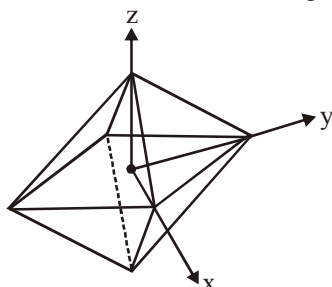
7.1 Limitations of VBT

1. The change in the properties of the ligands and the metal ions could not be explained.
2. The valence bond theory does not explain why certain complexes are more labile than other.
3. The VBT does not provide satisfactory explanation for the existence of inner orbital and outer orbital complexes.
4. The VBT could not explain the colour of complexes

8. CRYSTAL FIELD THEORY

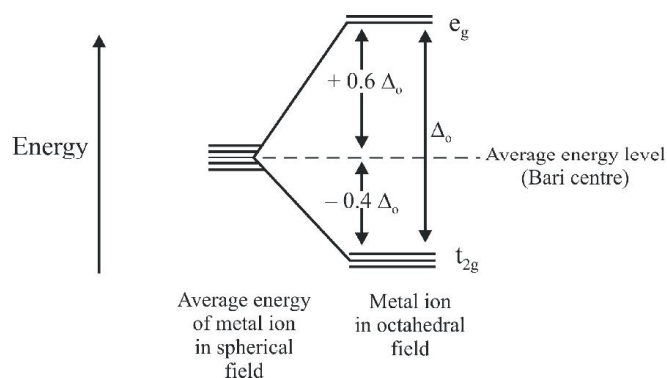
The Crystal Field Theory is more widely accepted than the valence bond theory. It assumes that the attraction between the central metal and the ligands in a complex is purely electrostatic. In the crystal field the following assumptions are made.

1. Ligands are treated as point charges.
2. There is no interaction between metal orbitals and ligand orbitals.
3. The d orbitals on the metal all have the same energy (that is degenerate) in the free atom. However, when a complex is formed the ligands destroy the degeneracy of these orbitals, i.e. the orbitals now have different energies.

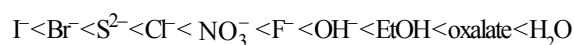
**8.1 Octahedral complexes**

In an octahedral complex, the metal is at the centre of the octahedron, and the ligands are at the six corners. The directions x , y and z point to three adjacent corners of the octahedron as shown.

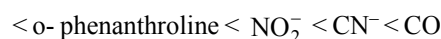
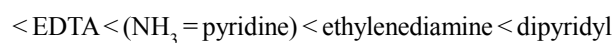
The lobes of the e_g orbitals ($d_{x^2-y^2}$ and d_{z^2}) point along the x , y and z axes. the lobes of the t_{2g} orbitals (d_{xy} , d_{xz} and d_{yz}) point in between the axes. The approach of six ligands along the x , y , z , $-x$, $-y$ and $-z$ directions will increase the energy of the $d_{x^2-y^2}$ and d_{z^2} orbitals (which point along the axes) much more than it increases the energy of the d_{xy} , d_{xz} and d_{yz} orbitals (which point between the axes). Thus under the influence of an octahedral ligand field the d orbitals split into two groups of different energies.



Ligands which cause only a small degree of crystal field splitting are termed weak field ligands. Ligands which cause a large splitting are called strong field ligands. The common ligands can be arranged in ascending order of crystal field splitting Δ .

Spectrochemical Series

(weak field ligands)



(strong field ligands)

A pattern of increasing σ donation is followed :

Halide donors < O donors < N donors < C donors

The total crystal field stabilization energy is given by

$$CFSE_{(\text{octahedral})} = -0.4n_{(t_{2g})} + 0.6n_{(e_g)}$$

where $n_{(t_{2g})}$ and $n_{(e_g)}$ are the number of electrons occupying the t_{2g} and e_g orbitals respectively. The CFSE is zero for ions with d^0 and d^{10} configurations in both strong and weak ligand field. The CFSE is also zero for d^5 configurations in a weak field.

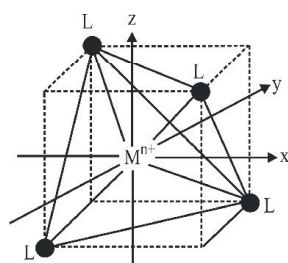
EFFECTS OF CRYSTAL FIELD SPLITTING

CFSE and electronic arrangements in octahedral complexes

Number of d electrons	Arrangement in weak ligand field				Arrangement in strong ligand field			
	t_{2g}	e_g	CFSE Δ_o	Spin only magnetic moment μ_s (D)	t_{2g}	e_g	CFSE Δ_o	Spin only magnetic moment μ_s (D)
d^1	$\uparrow$		-0.4	1.73	$\uparrow$		-0.4	1.73
d^2	$\uparrow\uparrow$		-0.8	2.83	$\uparrow\uparrow$		-0.8	2.83
d^3	$\uparrow\uparrow\uparrow$		-1.2	3.87	$\uparrow\uparrow\uparrow$		-1.2	3.87
d^4	$\uparrow\uparrow\uparrow$	$\uparrow$	-0.6	4.90	$\uparrow\downarrow\uparrow\uparrow$		-1.6	2.83
d^5	$\uparrow\uparrow\uparrow$	$\uparrow\uparrow$	-0.0	5.92	$\uparrow\downarrow\uparrow\downarrow\uparrow$		-2.0	1.73
d^6	$\uparrow\downarrow\uparrow\uparrow$	$\uparrow\uparrow$	-0.4	4.90	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$		-2.4	0.00
d^7	$\uparrow\downarrow\uparrow\downarrow\uparrow$	$\uparrow\uparrow$	-0.8	3.87	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$	$\uparrow$	-1.8	1.73
d^8	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$	$\uparrow\uparrow$	-1.2	2.83	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$	$\uparrow\uparrow$	-1.2	2.83
d^9	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$	$\uparrow\downarrow\uparrow$	-0.6	1.73	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$	$\uparrow\downarrow\uparrow$	-0.6	1.73
d^{10}	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$	$\uparrow\downarrow\uparrow\downarrow$	0.0	0.00	$\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow$	$\uparrow\downarrow\uparrow\downarrow$	0.0	0.00

8.2 Tetrahedral Complexes

A regular tetrahedron is related to a cube. One atom is at the centre of the cube, and four of the eight corners of the cube are occupied by ligands as shown.



Tetrahedral arrangement of four ligands (L) around the metal ion (M^{n+}) in tetrahedral complex ion, (ML_4) $^{n+}$

The directions x, y and z point to the centres of the faces of the cube. The e orbitals point along x, y and z axes (that is to the centres of the faces). The t_2 orbitals point between x, y and z axes (that is towards the centres of the edges of the cube). The direction of approach of the ligands does not coincide exactly with either the e or the t_2 orbitals.

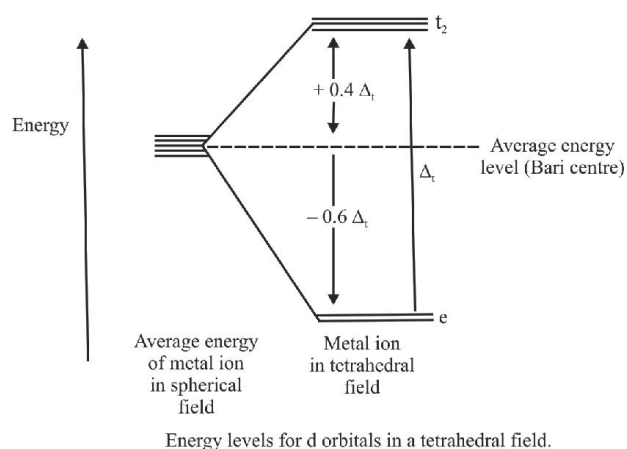
Thus the t_2 orbitals are nearer to the direction of the ligands than the e orbitals. The approach of the ligands raises the energy of both sets of orbitals. The energy of the t_2 orbitals is raised most because they are closest to the ligands. The crystal field splitting is the opposite way round to that in octahedral complexes

The t_2 orbitals are $0.4\Delta_t$ above weighted average energy of the two groups (the bary centre) and the e orbitals are $0.6\Delta_t$ below the average.

The magnitude of the crystal field splitting Δ_t in tetrahedral complexes is considerably less than in octahedral fields. There are two reasons for this :

1. There are only four ligands instead of six, so the ligand field is only two third the size ; hence the ligand field splitting is also two third the size.
2. The direction of the orbitals does not coincide with the direction of the ligands. This reduces the crystal field splitting by roughly a further two third.

Thus the tetrahedral crystal field splitting Δ_t is roughly $2/3 \times 2/3 = 4/9$ of the octahedral crystal field splitting Δ_o .



9. ORGANOMETALLIC COMPOUNDS

Compounds that contain at least one carbon-metal bond are called organometallic compounds.

Grignard reagent, RMgX is a familiar example of organometallic compounds where R is an alkyl group. Diethyl zinc $[\text{Zn}(\text{C}_2\text{H}_5)_2]$, lead tetraethyl $[\text{Pb}(\text{C}_2\text{H}_5)_4]$, ferrocene $[\text{Fe}(\text{C}_5\text{H}_5)_2]$, dibenzene chromium $[\text{Cr}(\text{C}_6\text{H}_6)_2]$, metal carbonyls are other examples of organometallic compounds.

Organometallic compounds may be classified in three classes :

1. Sigma (σ) bonded complexes.
2. Pi (π) bonded complexes,
3. Complexes containing both σ - and π -bonding characteristics.

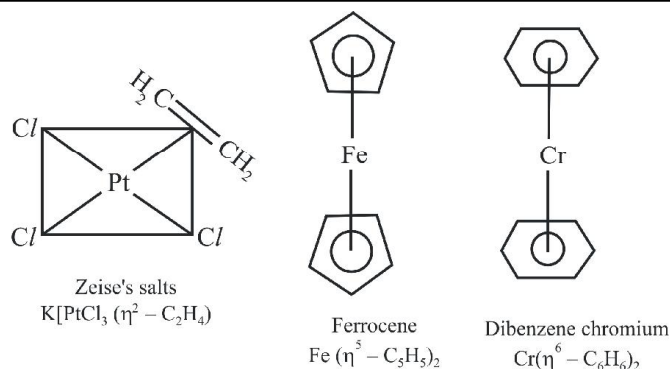
9.1 Sigma bonded complexes

In these complexes, the metal atom and carbon atom of the ligand are joined together with a sigma bond, i.e., the ligand contributes one electron and is, therefore, called one electron donor. Examples are :

- (i) Grignard reagent, R-Mg-X where R is an alkyl or aryl group and X is halogen.
- (ii) Zinc compounds of the formula R_2Zn such as $(\text{C}_2\text{H}_5)_2\text{Zn}$. This was first isolated by Frankland in 1849. Other similar compounds are $(\text{CH}_3)_4\text{Sn}$, $(\text{C}_2\text{H}_5)_4\text{Pb}$, $\text{Al}_2(\text{CH}_3)_6$, $\text{Al}_2(\text{C}_2\text{H}_5)_6$ and $\text{Pb}(\text{CH}_3)_4$, etc.

9.2 π -bonded organometallic compounds

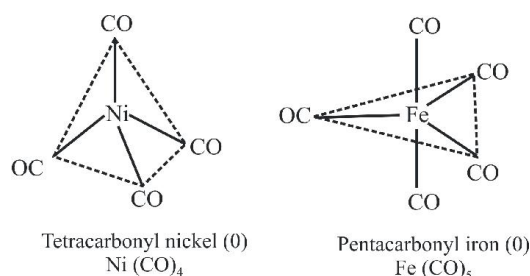
These are the compounds of metals with alkenes, alkynes, benzene and other ring compounds. In these complexes, the metal and ligand form a bond that involves the π electrons of the ligand. Three common examples are Zeise's salt, ferrocene and dibenzene chromium. These are shown here :



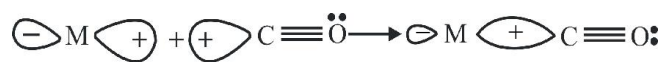
The number of carbon atoms bound to the metal in these compounds is indicated by the Greek letter ' η ' (eta) with a number. The prefixes η^2 , η^5 and η^6 indicate that 2, 5 and 6 carbon atoms are bound to the metal in the compound.

9.3 σ - and π -bonded organometallic compounds

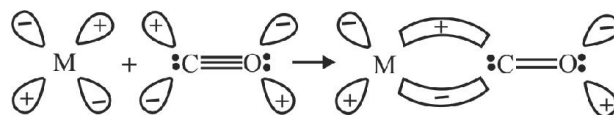
Metal carbonyls, compounds formed between metal and carbon monoxide belong to this class. These compounds possess both σ - and π -bonding. The oxidation state of metal atoms in these compounds is zero. Carbonyls may be monomeric, bridged or polynuclear.



In a metal carbonyl, the metal-carbon bond possesses both the σ - and π -character. A σ -bond between metal and carbon atom is formed when a vacant hybrid orbital of the metal atom overlaps with an orbital on C atom of carbon monoxide containing a lone pair of electrons.



Formation of π -bond is caused when a filled orbital of the metal atom overlaps with a vacant antibonding π^* orbital of C atom of carbon monoxide. This overlap is also called back donation of electrons by metal atom to carbon. It has been shown below :

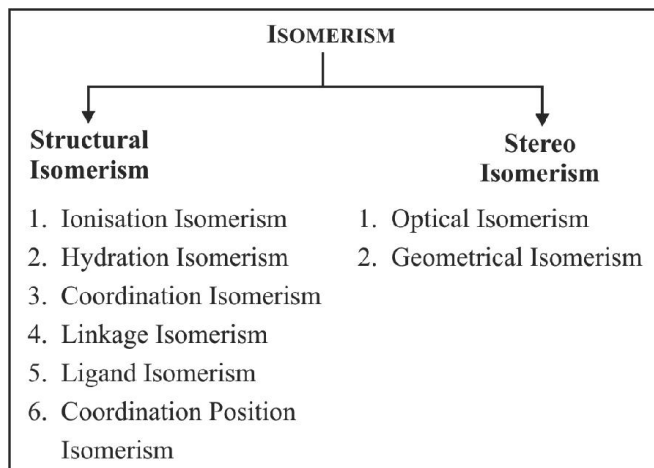


The π -overlap is perpendicular to the nodal plane of σ -bond.

In olefinic complexes, the bonding π -orbital electrons are donated to the empty orbital of the metal atom and at the same time back bonding occurs from filled orbital of the metal atom to the antibonding π -orbital of the olefin.

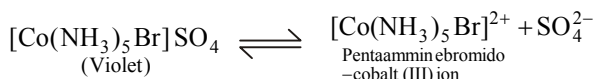
10. ISOMERISM

The compounds having same molecular formula but different structural formula are called isomers.

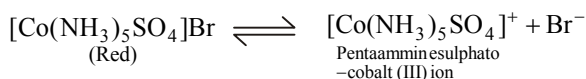


10.1 Structural Isomerism

10.1.1 Ionisation Isomerism : This type of isomerism arises when the coordination compounds give different ions in solution. For example, there are two isomers of the formula $\text{Co}(\text{NH}_3)_5\text{BrSO}_4$.



This isomer gives a white precipitate of BaSO_4 with BaCl_2 solution.

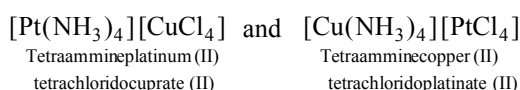


Above isomer gives light yellow precipitate with AgNO_3 solution.

10.1.2 Hydrate Isomerism : This type of isomerism arises when different number of water molecules are present inside and outside the coordination sphere. This isomerism is best illustrated by the three isomers that have the formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$.

$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$, and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ are its Hydrate Isomers.

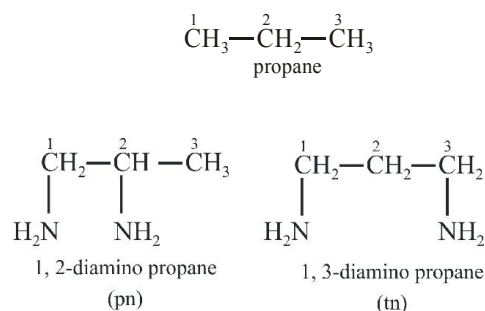
10.1.3 Coordination Isomerism : This type of isomerism is observed in the coordination compounds having both cationic and anionic complex ions. The ligands are interchanged in both the cationic and anionic ions to form isomers. An examples is :



10.1.4 Linkage Isomerism : This type of isomerism occurs in complex compounds which contain ambidentate ligands like NO_2^- , SCN^- , CN^- , $\text{S}_2\text{O}_3^{2-}$ and CO .

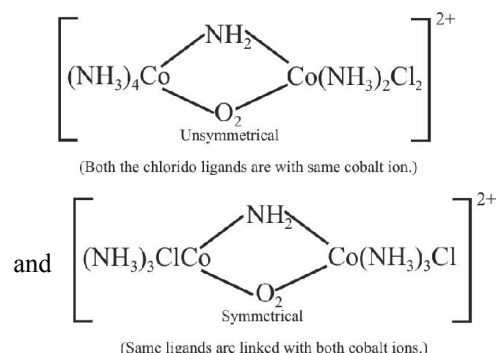
For example, $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$ are linkage isomers as NO_2^- is linked through N or through O.

10.1.5 Ligand Isomerism : Some ligands themselves are capable of existing as isomers, e.g., diamino propane can exist both as 1, 2-diamino propane (pn) and 1, 3-diamino propane, also called trimethylene diamine (tn).



When these ligands (i.e., pn and tn) are associated into complexes the complexes are isomers of each other. One example of isomeric complexes having this ligand is : $[\text{Co}(\text{pn})_2\text{Cl}_2]^+$ and $[\text{Co}(\text{tn})_2\text{Cl}_2]^+$ ions.

10.1.6 Coordination position isomerism : This type of isomerism is exhibited by polynuclear complexes by changing the position of ligands with respect to different metal atoms present in the complex. For example,



10.2 Stereo Isomerism

Stereo isomerism is exhibited by those compounds which have the same position of atoms or groups but these atoms or groups have different arrangement around the central atom.

10.2.1 Geometrical Isomerism

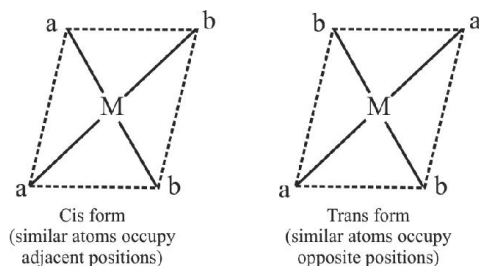
The complex compounds which have the same ligands in the co-ordination sphere but the relative position of the

ligands around the central metal atom is different are called geometrical isomers and the phenomenon is called geometrical isomerism.

10.2.1.1 Geometrical Isomerism in square planar complexes

A square planar complex having similar ligands at adjacent positions (90° apart) is called cis - isomer while a square planar complex having two similar ligands at opposite positions (180° apart) is called trans-isomer.

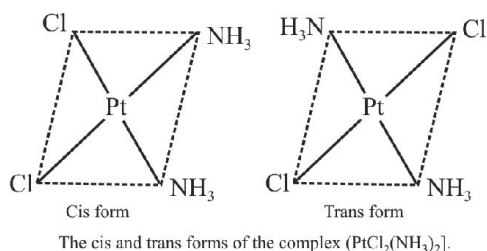
1. Ma_2b_2



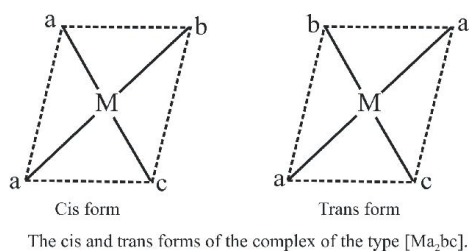
Example - 1

Draw the geometrical isomers of $[PtCl_2(NH_3)_2]$

Sol.



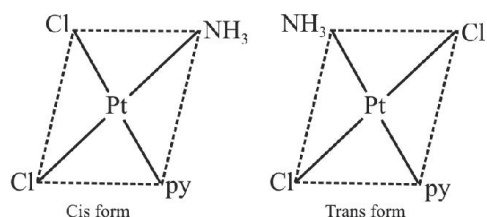
2. Ma_2bc



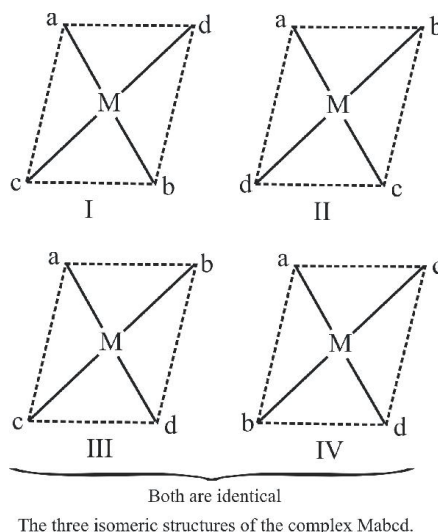
Example - 2

Draw the geometrical isomers of $[PtCl_2(NH_3)py]$

Sol.



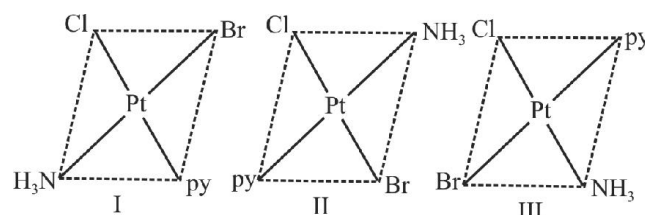
3. $Mabcd$



Example - 3

Draw the geometrical isomers of $[PtClBrpy(NH_3)]$

Sol.

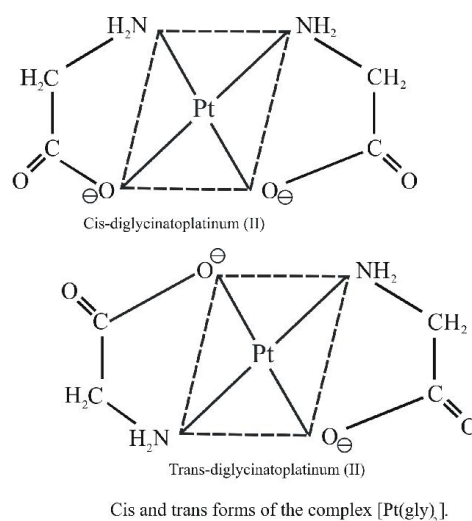


4. $M(AB)_2$

Example - 4

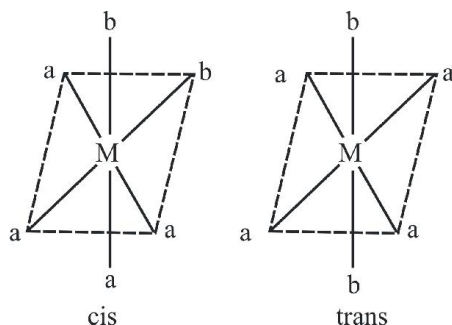
Draw the geometrical isomers of $[Pt(gly)_2]$

Sol.



10.2.1.2 Geometrical Isomerism in octahedral complexes

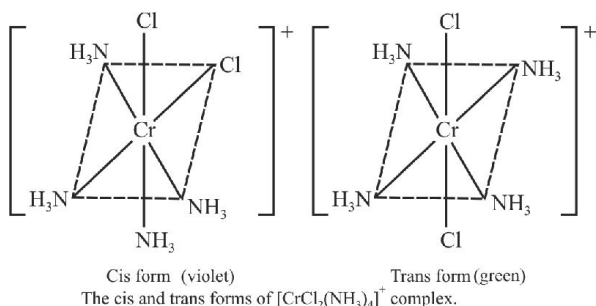
1. Ma_4b_2



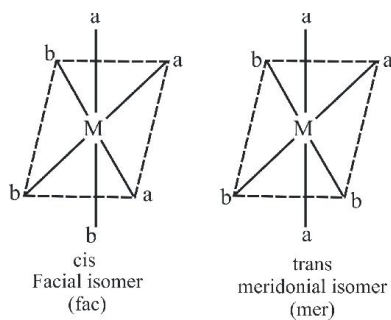
Example - 5

Draw the geometrical isomers of $[\text{CrCl}_2(\text{NH}_3)_4]^+$

Sol.



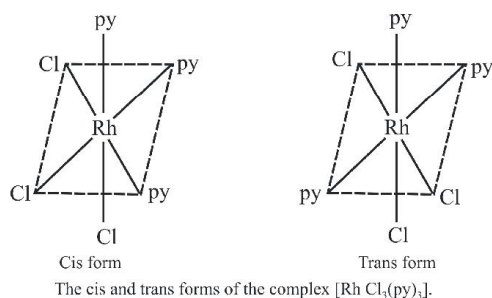
2. Ma_3b_3



Example - 6

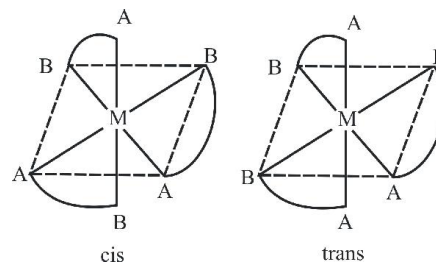
Draw the geometrical isomers of $[\text{RhCl}_3(\text{py})_3]$

Sol.



3. Mabcdef : They form 15 isomers

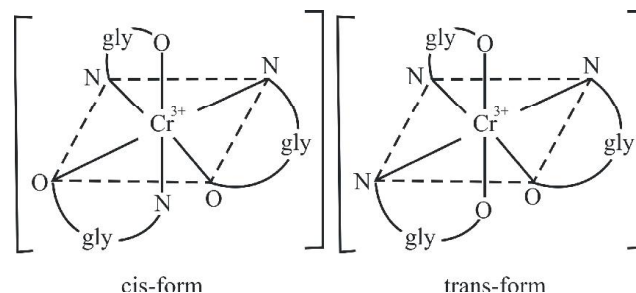
4. $\text{M}(\text{AB})_3$



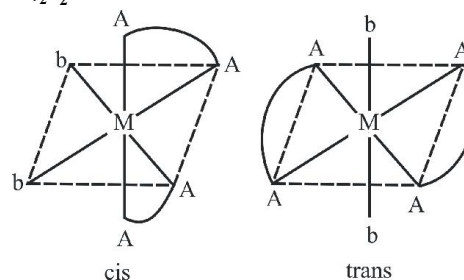
Example - 7

Draw the geometrical isomers of $[\text{Cr}(\text{gly})_3]$

Sol.



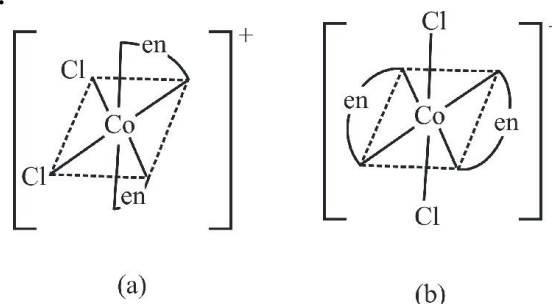
5. $\text{M}(\text{AA})_2\text{b}_2$



Example - 8

Draw the geometrical isomers of $[\text{CoCl}_2(\text{en})_2]^+$

Sol.



Cis and trans-isomers of $[\text{Co}^{\text{III}}(\text{en})_2\text{Cl}_2]^+$ ion.

(a) Cis-isomer

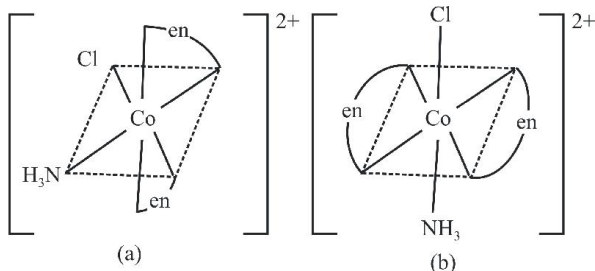
(b) trans-isomer

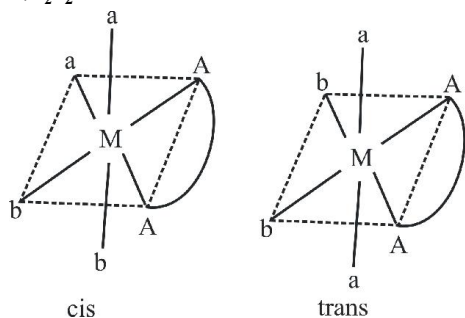
6. $M(AA)_2bc$

Example - 9

Draw the geometrical isomers of $[Co^{III}(en)_2(NH_3)(Cl)]^{2+}$

Sol.

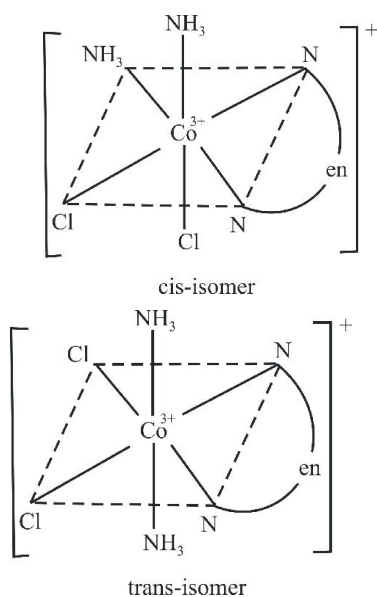

Cis and trans-isomers $[Co^{III}(en)_2(NH_3)(Cl)]^{2+}$ ion

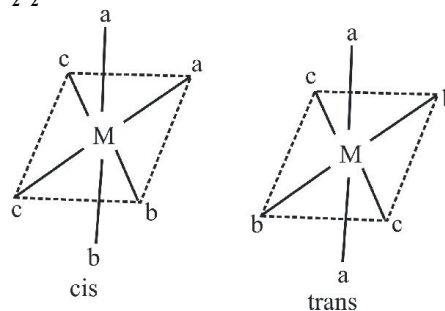
7. $M(AA)_2b_2$


Example - 10

Draw the geometrical isomers of $[Co^{III}(en)(NH_3)_2Cl_2]^+$

Sol.

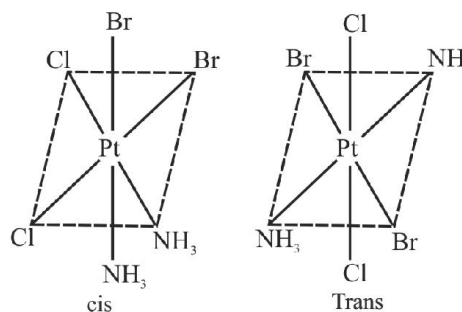

cis and trans-isomers of $[Co^{III}(en)(NH_3)_2Cl_2]^+$

8. $Ma_2b_2c_2$


Example - 11

Draw the geometrical isomers of $[PtCl_2Br_2(NH_3)_2]$

Sol.



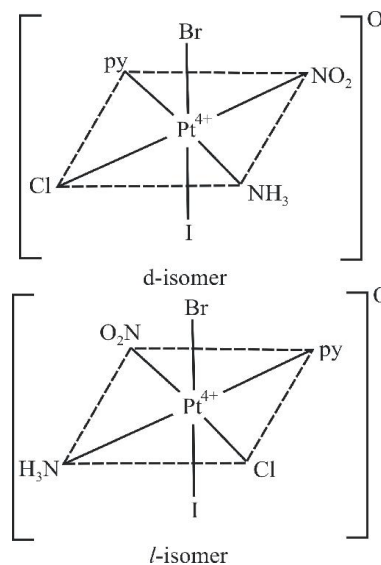
Optical Isomerism in octahedral complexes

1. $Mabcdef$

Example - 12

Draw the optical isomers of $[Pt^{IV}(Cl)(Br)(I)(py)(NO_2)(NH_3)]$

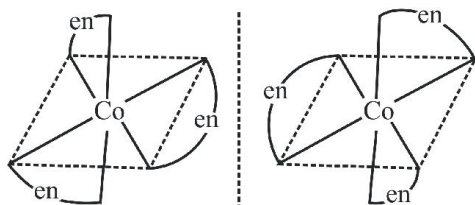
Sol.


Optical d and l-forms $[Pt^{IV}(py)(NH_3)(NO_2)(Cl)(Br)(I)]^0$

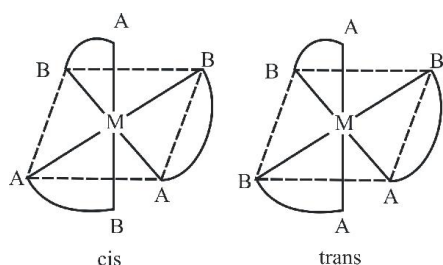
2. $M(AA)_3$
Example - 13

Draw the optical isomers of $[Co(en)_3]^{3+}$

Sol.

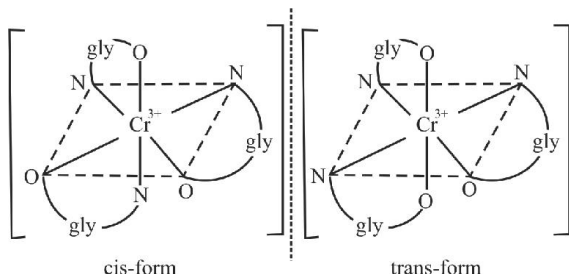


The two optical isomeric forms of the complex $[Co(en)_3]^{3+}$

3. $M(AB)_3$

Example - 14

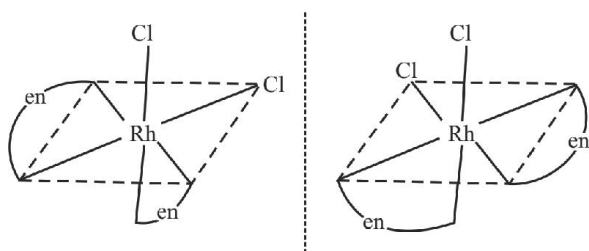
Draw the optical isomers of $[Cr(gly)_3]$

Sol.

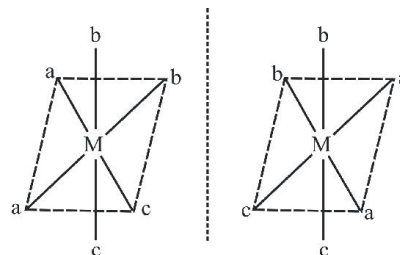

4. $cis M(AA)_2b_2$
Example - 15

Draw the optical isomers of $[RhCl_2(en)_2]^+$

Sol.

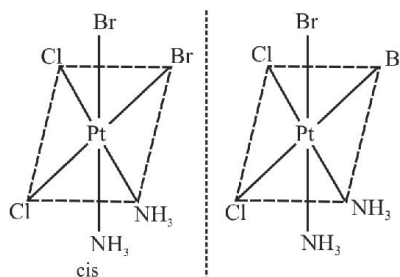


Optical active isomers of $cis [RhCl_2(en)_2]^+$

5. $cis Ma_2b_2c_2$

Example - 16

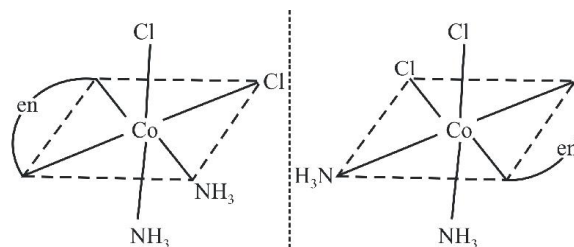
Draw the optical isomers of $[PtCl_2Br_2(NH_3)_2]$

Sol.

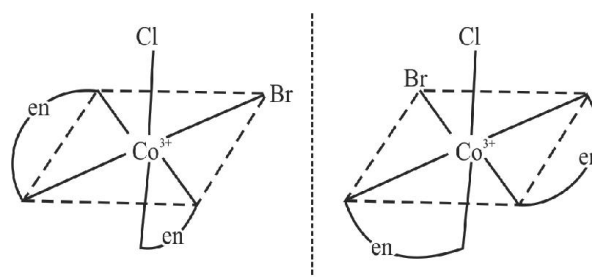

6. $cis M(AA)b_2c_2$
Example - 17

Draw the optical isomers of $[CoCl_2(en)(NH_3)_2]^+$

Sol.


7. $cis M(AA)_2bc$
Example - 18

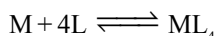
Draw the optical isomers of $[CoCl(en)_2Br]^{2+}$



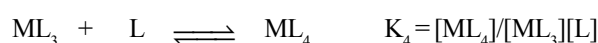
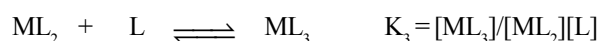
11. STABILITY OF COORDINATION COMPOUNDS

The stability of a complex in solution refers to the degree of association between the two species involved in the state of equilibrium.

If we have a reaction of the type :



then the larger the stability constant, the higher the proportion of ML_4 that exists in solution. Free metal ions rarely exist in the solution so that M will usually be surrounded by solvent molecules which will compete with the ligand molecules, L, and be successively replaced by them. For simplicity, we generally ignore these solvent molecules and write four stability constants as follows :



where K_1, K_2 , etc., are referred to as stepwise stability constants. Alternatively, we can write the overall stability constant thus :



The stepwise and overall stability constant are therefore related as follows :

$$\beta_4 = K_1 \times K_2 \times K_3 \times K_4 \text{ or more generally,}$$

$$\beta_n = K_1 \times K_2 \times K_3 \times K_4 \dots K_n$$

The instability constant or the dissociation constant of coordination compounds is defined as the reciprocal of the formation constant.

12. IMPORTANCE AND APPLICATIONS OF COORDINATION COMPOUNDS

- Analytical chemistry :** The analytical applications of coordination chemistry are in
 - Qualitative and quantitative analysis :** Metals ions form colored coordination compounds on reaction with a number of ligands. These reactions are used for detection of the metal ions. The colored complexes formed can be used for the estimation of metals by classical or instrumental methods such as gravimetry or colorimetry. Some examples are given as follows :

The presence of iron ions (Fe^{3+}) can be detected by the addition potassium ferrocyanide solution, which results in formation of Prussian blue complex.

$$Fe^{2+} + K_3Fe(CN)_6 \rightarrow KFe[Fe(CN)_6] + 2K^+$$
 - Volumetric analysis :** Hardness of water can be estimated by titration with EDTA. The metal ions causing hardness, that is Ca^{2+} and Mg^{2+} , form stable complexes with EDTA.
- Metal extraction and purification :** Extraction of metals, such as silver and gold, is carried out by forming their water soluble cyanide complexes with the ore. Pure gold can then be obtained from the solution by addition of zinc. Similarly, metals can be purified by formation and then decomposition of their coordination compounds. For example, impure nickel obtained after extraction may be converted into pure nickel by first converting it to nickel carbonyl and then decomposing.
- Catalysis :** Coordination compounds are used as catalysts in important commercial processes. For example,
 - The Zeigler-Natta catalyst ($TiCl_4$ and trialkyl aluminium) is used as a catalyst in the formation of polyethylene.
 - The Wilkinson catalyst - $RhCl(PPh_3)_3$ is used in the hydrogenation of alkenes.
 - In the Monsanto acetic acid process, various rhodium complexes, such as $[Rh(CO)_2I_2]$, $[Rh(Cl)(CO)(PPh_3)_2]$ or $[Rh(Cl)(CO)_2I_2]$, are used as catalyst in the presence of CH_3I , I_2 or HI as activator.
- Electroplating :** Coordination compounds of gold, silver and copper are used as components in the baths used for electroplating articles of other metals with these metals. For example, in silver plating, $K[Ag(CN)_2]$ is used as an electrolyte; in gold plating, $K[Au(CN)_2]$ is used as an electrolyte; and in copper plating, $K_3[Cu(CN)_4]$ is used as an electrolyte.
- Biological importance :** Some important biological compounds are coordination complexes. For example, chlorophyll is a complex of Mg^{2+} . This green pigment plays a vital role in photosynthesis in plants. Similarly, haemoglobin, the red pigment present in blood, is a coordination complex of Fe^{2+} and vitamin B_{12} , an essential nutrient, is a complex compound of Co^{3+} .
- Medicinal uses :** Complexing or chelating agents are used in treating metal poisoning, wherein, the coordination complex is formed between toxic metal in excess metal and the complexing agent. For example, EDTA is used in lead poisoning. EDTA, when injected intravenously into the bloodstream, traps lead forming a compound that is flushed out of the body with the urine. Other heavy metal poisonings that can be treated similarly with chelation therapy are mercury, arsenic, aluminium, chromium, cobalt, manganese, nickel, selenium, zinc, tin and thallium. Similarly, chelating ligands D-penicillamine and desferrioxime B are used for removal of excess copper and iron, respectively. New potent drugs are being created using various derivatives of metallocene. A platinum complex $[PtCl_2(NH_3)_2]$ called cis-platin is used in treatment of cancer.

13. CO-ORDINATION COMPOUNDS AND COMPLEX IONS

- (a) Co-ordination compounds are the compounds in which the central metal atom is linked to a number of ligands (ions or neutral molecules) by co-ordinate bonds i.e. by donation of lone pairs of electrons by these ligands to the central metal atom ion.

If a such a compound carries positive or negative charge, it is called a complex ion, e.g. $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$. Hence Co-ordination compounds may also be defined as those compounds which contain complex ions, e.g., $\text{K}_4[\text{Fe}(\text{CN})_6]$, $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$, etc. In general, a complex ion is represented as $[\text{ML}_n]^{\pm x}$. Where M is the metal ion, L represents ligands, n is the coordination number of metal ion and x is the net charge on the complex.

- (b) There are four types of complexes :
- (i) Cation as complex ion, (carrying a net positive charge) e.g., $[\text{Cr}(\text{NH}_3)_6]^{3+}$ in $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$.
- (ii) Anion as complex ion, (carrying a net negative charge) e.g., $[\text{Fe}(\text{CN})_6]^{3-}$ in $\text{K}_3[\text{Fe}(\text{CN})_6]$.
- (iii) Cation and anion both as complex ion. Carrying both +ive and -ive change. For e.g., $[\text{Pt}(\text{Py})_4][\text{PtCl}_4]$.
- (iv) Neutral complex (A complex carrying no net charge) e.g., $[\text{Ni}(\text{CO})_4]$ etc.

14. TERMINOLOGY OF CO-ORDINATION COMPOUNDS

14.1 Centre of Coordination

(Central atom/ion or Acceptor atom/ion) :

The cation or neutral atom to which one or more ligands (neutral molecules or anions) are attached or coordinated is the centre of coordination.

The central atom/ion must have empty orbitals as it acts as an acceptor and thus has to accommodate electron pairs donated by the donor atom of the ligand. This explains why the transition metal having empty d-orbitals, form coordination compounds very readily.

For example in the complexes $[\text{Ni}(\text{NH}_3)_6]^{2+}$ and $[(\text{CN})_6]^{3-}$, Ni^{2+} and Fe^{3+} respectively are the central ions.

14.2 Ligands

An atom, ion or molecule which can donate atleast a pair of electrons to the central atom to form a coordinate bond (or dative linkage) is called as a ligand or a coordinating group. Further in a ligand, the particular atom which actually donates the electron pair is called the donor atom.

The ligands act as Lewis bases as they donate one or more electron pair to the central metal atoms or ions which act as Lewis acids by accepting electrons.

Example

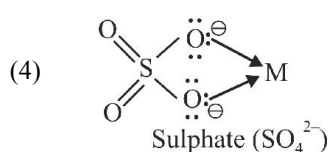
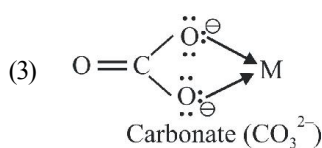
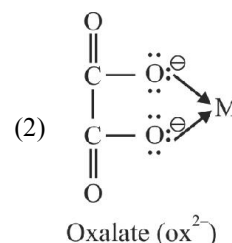
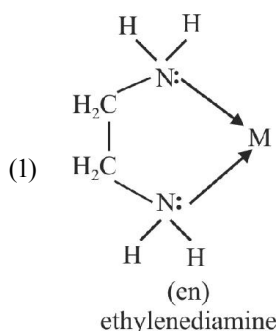


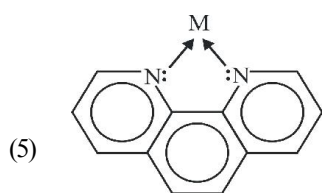
14.2.1 Types of Ligands : Ligands can be classified on the number of lone pair electrons they donate to the central metal atom or ion.

- (a) **Monodentate or unidentate ligands :** They have one donor atom that donates only one electron pair to central metal atom. eg : F^- , Cl^- , Br^- , H_2O , NH_3 , CN^- , NO_2^- , OH^- , NH_2^- , CO , R-OH , SCN^- , $\text{R}-\ddot{\text{O}}-\text{R}$, I^- , SH^- , RCOO^- , RS^- , $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{S}}-\text{CH}_3$ (Dimethyl sulfoxide), NH_2 , NH_2^- , NO_2^- , NO_3^- , S^{2-} , O^{2-} , pyridine.

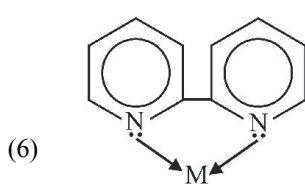
- (b) **Bidentate ligands :** Ligands which have two donor atoms and have the ability to link with central metal at two positions are called bidentate ligands

Example

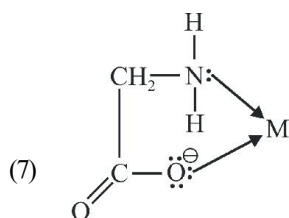
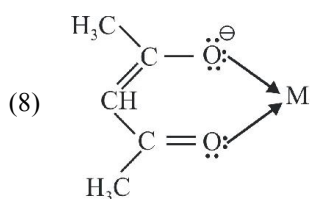
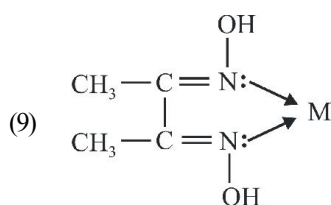
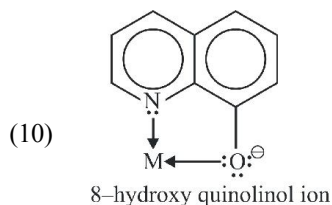




1,10 – Phenanthroline (Phen)



2, 2 – Dipyridyl (Dipy)

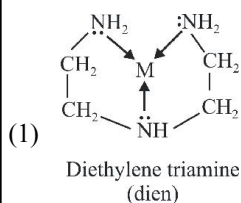
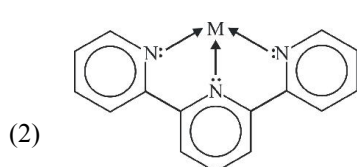

glycinate (gly⁻)

acetyl acetone (acac⁻)

Dimethyl glyoxime
(dmg)


8-hydroxy quinolinol ion

(c) Tridentate Ligand

The ligands having three donor atoms are called tridentate ligands.

Example :

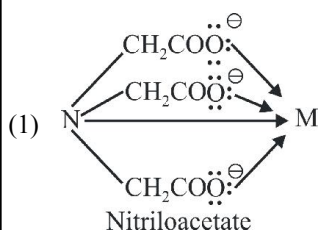

Diethylene triamine
(dien)


2,2',2''-Terpyridine (terpy)

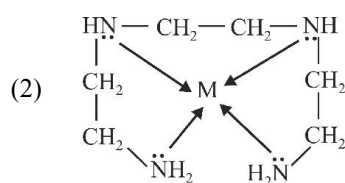
(d) Tetradentate ligand

These ligand possess four donor atoms

Example :



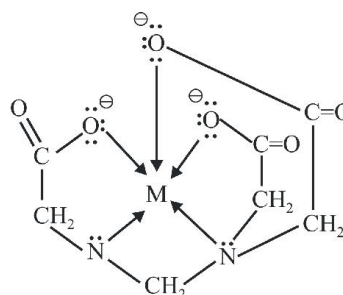
Nitriloacetate


Triethylene tetramine
(trien)

(e) Pentadentate ligands

They have five donor atoms

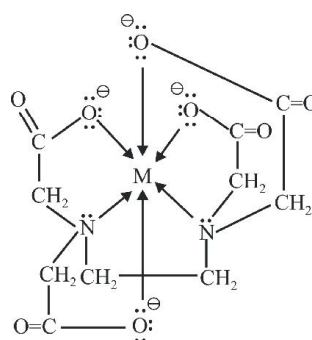
Example :


Ethylenediamine
triacetate ion

(f) Hexadentate Ligands

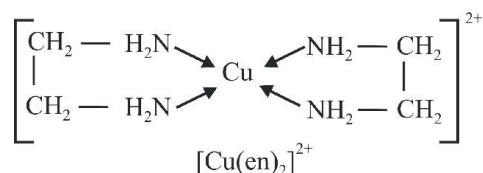
They have six donor atoms.

Example :


Ethylenediamine
Tetracetate ion
(EDTA)⁴⁻

14.2.2 Chelating ligands :

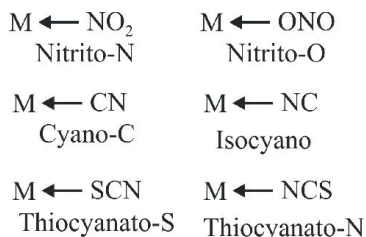
A bidentate or a polydentate ligand is known as a chelating ligand if on co-ordination it results in the formation of a cyclic ring structure. The complex thus formed are called chelates.



The chelates containing 5 or 6 membered rings are more stable. Ligands with larger groups form more unstable rings than with smaller groups due to steric hinderance.

14.2.3 Ambidentate ligands :

The ligands which have two donor atoms but in forming complexes only one donor atom is attached to the metal atom at a given time. Such ligands are called ambidentate ligands.

Example :**15. CO-ORDINATION NUMBER (C.N)**

The number of atoms of the ligands that directly bound to the central metal atom or ion by co-ordinate bonds is known as the co-ordination number of the metal atom or ion. It is also equal to the secondary valency.

Complex	Co-ordination numbers
$K_4 [Fe(CN)_6]$	6
$[Ag(CN)_2]^-$	2
$[Pt(NH_3)_2 Cl_2]$	4
$[Ca(EDTA)]^{2-}$	6

16. STABILITY OF COORDINATION COMPOUNDS IN SOLUTIONS

16.1 In general, higher the charge density on the central ion, the greater is the stability of its complexes, i.e., the higher value

of $\frac{\text{charge}}{\text{radius of the ion}}$, the greater is the stability of its

complexes. Electronegativity of the central ion influences the stability. The higher the electronegativity of the central ion, the greater is the stability of its complexes.

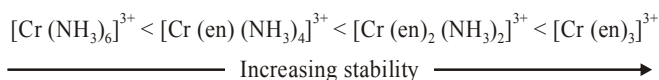
16.2. The higher the oxidation state of the metal, the more stable is the complex. The charge density of Co^{3+} ion is more than Co^{2+} ion and thus, $[Co(NH_3)_6]^{3+}$ is more stable than $[Co(NH_3)_6]^{2+}$. Similarly, $[Fe(CN)_6]^{3-}$ is more stable than $[Fe(CN)_6]^{4-}$.

The cyano and ammine complexes are far more stable than those formed by halide ions. This is due to the fact that NH_3 and CN^- are strong Lewis bases.

The complexes of bivalent cations (M^{2+}) of 3d-series shown the following order of stability :

Cation	Mn^{2+}	Fe^{2+}	Co^{2+}	Ni^{2+}	Cu^{2+}	
Ionic size	0.91	0.83	0.82	0.78	0.69	decreases
Stability of the complex						increases (Irving William order)

Chelating ligands form more stable complexes as compared to monodentate ligands. Greater is the chelation, more is the stability of complex.



SOLVED EXAMPLES

Example - 1

- (a) Name one substance which can act as both
 (i) Analgesic and antipyretic
 (ii) Antiseptic and disinfectant
 (b) Explain the following terms with suitable example of each
 (i) Broad spectrum antibiotics
 (ii) Anionic detergents

- Sol.** (a) (i) Aspirin, (ii) Phenol
 (b) Antibiotics which kill or inhibit the growth of wide range of gram positive and gram negative bacteria e.g., Chloramphenicol.
 (c) Sodium salts of sulphonated long chain alcohols or hydrocarbons are anionic detergents, e.g., Sodium lauryl sulphate.

Example - 2

- (a) Give the electronic configuration of the d-orbitals of Ti in $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ ion and explain why this complex is coloured? [At. No. of Ti = 22]
 (b) Write IUPAC name of $[\text{Cr}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Cl}_3$

- Sol.** (a) $\text{Ti}^{3+} = 3d^1 4s^0$
 $\therefore$ Due to d-transition, the complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is coloured.
 (b) IUPAC name is Triamminetriaquachromium (III) Chloride.

Example - 3

Among Octahedral and Tetrahedral crystal fields, in which case the magnitude of crystal field splitting is larger?

Sol. Octahedral crystal field.

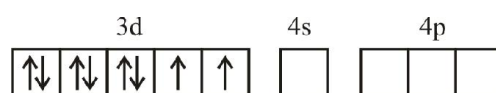
Example - 4

- Describe the followings :
 (i) Optical isomerism
 (ii) Magnetic behaviour of $[\text{Ni}(\text{CN})_4]^{2-}$ ion. (At. No. of Ni = 28)
 (iii) Preparation of tetrabutyl lead tin

Sol. (i) Optical isomerism is shown by compounds having chiral molecules, i.e., molecules which do not have plane of symmetry. These isomers are non-

superimposable mirror images of each other. They are optically active and rotate the plane of polarisation equally but in opposite directions. The isomer which rotated the plane of polarised light towards left is called laevorotatory (-1) while the isomer which rotates plane towards right is called dextro-rotatory (+).

(ii) Atomic orbitals of Ni (II) ion



dsp^2 hybridisation in Ni (II) ion



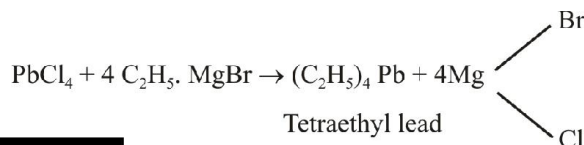
$[\text{Ni}(\text{CN})_4]^{2-}$ ion



Four electron pairs form CN^- groups

As there are no unpaired electrons, the $[\text{Ni}(\text{CN})_4]^{2-}$ ion is diamagnetic.

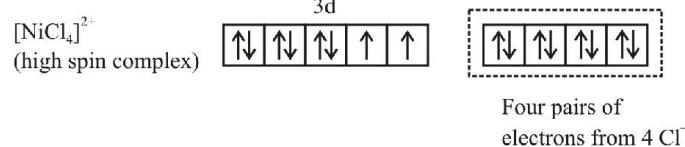
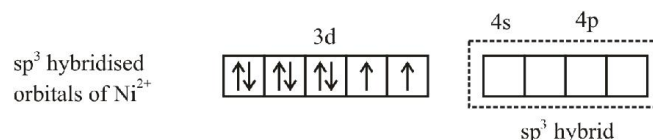
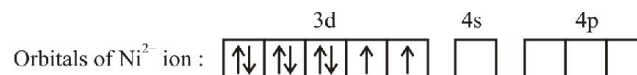
(iii) Tetraethyl lead is prepared by reaction of PbCl_4 with ethyl magnesium bromide.



Example - 5

Using valence bond theory, predict the geometry and magnetic character of $[\text{NiCl}_4]^{2-}$. Atomic no. of Ni is 28.

Sol.



Geometry : Tetrahedral due to sp^3 hybridisation.

Magnetic character : Paramagnetic, due to the presence of 2 unpaired electron in 3d-orbitals.

Example - 6

- (a) Give the IUPAC name of $[\text{PtCl}(\text{NH}_2\text{CH}_3)_2(\text{NH}_3)_2]\text{Cl}$
 (b) Write the name of linkage isomer of $[\text{Co}(\text{ONO})(\text{NH}_3)_5]^{2+}$
 (c) Though CO is a weak lewis base yet it forms a number of stable metal carbonyls. Explain.

Sol. (a) Diamminechloro bis (methylamine) platinum (II) chloride.
 (b) Pentaamine nitrito – N – cobalt (III) cation.
 (c) CO is a weak electron donar and still metal carbonyls are stable. This is because of back bonding resulting from delocalisation of electrons from filled d-orbitals of metal into the empty antibonding molecular orbitals of CO ligands.

Example - 7

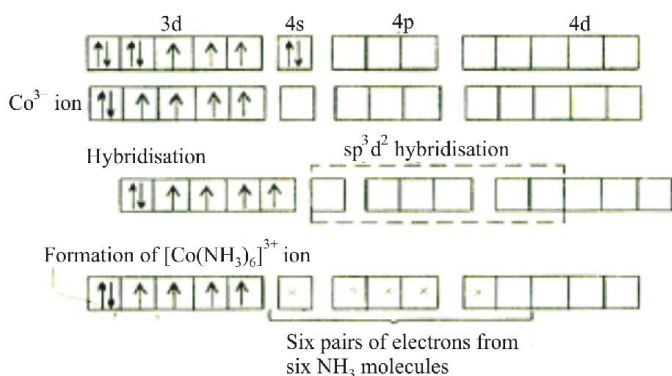
Write the IUPAC name for any of the isomers with the molecular formula $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]\text{Cl}_2$

Sol. Diamine dichloro platinum (IV) chloride.

Example - 8

Using the valence bond approach predict the shape and magnetic character of $[\text{Co}(\text{NH}_3)_6]^{3+}$. (Atomic number of Co is 27)

Sol. In $[\text{Co}(\text{NH}_3)_6]^{3+}$, the oxidation state of Co is +3.
 Co (Z = 27) atom is ground state



Due to sp^3d^2 hybridisation, the complex ion has octahedral shape. Due to the presence of four unpaired electrons, the complex ion is paramagnetic.

Example - 9

Write the IUPAC name for any one of the isomers of $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$.

Sol. Tetra ammine dichloro chromium (III) ion.

Example - 10

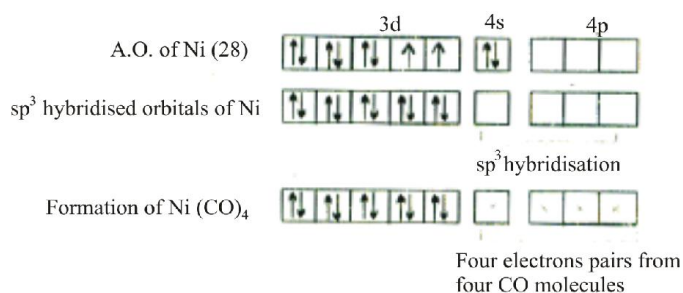
What type of hybrid orbitals are associated with Ni atom in $[\text{Ni}(\text{CN})_4]^{2-}$

Sol. dsp^2

Example - 11

Using the valence bond approach, predict the shape and magnetic character of $[\text{Ni}(\text{CO})_4]$. (At. No. of Ni = 28).

Sol. In the complex $[\text{Ni}(\text{CO})_4]$, the oxidation state of Ni is '0'. Its electronic configurations is $[\text{Ar}] 3\text{d}^8 4\text{s}^2$.



The resulting complex has tetrahedral shape and is diamagnetic due to absence of unpaired electrons.

Example - 12

What is a bidentate ligand ?

Sol. A ligand which contains two donor atoms so that it can form two coordinate bonds with the central metal atom.

Example : Ethylene diammine : $\text{NH}_2 - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$

Example - 13

What is the coordination number of central metal ion in $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{2-}$?

Sol. Six

Example - 14

Explain each of the following observations :

(i) Tetrahedral Ni

(ii) Complexes are paramagnetic but square planar Ni

Sol. (i) In the complex $[\text{NiCl}_4]^{2-}$, nickel undergoes sp^3 hybridisation and the complex has tetrahedral geometry. In the presence of weak ligand field Cl^- , the unpaired electrons of nickel do not paired up. So the complex is paramagnetic.

In complex like, $[\text{Ni}(\text{CN})_4]^{2-}$, the unpaired electrons of 3d orbitals are forced to pair in the presence of strong ligand field of CN^- , Ni^{2+} ion undergoes dsp^2 hybridisation resulting in square planar geometry. As there are no unpaired electrons, the complex is diamagnetic.

(ii) π -complex are the compounds of transition metals with alkenes, alkynes, benzene and other ring system. In such compounds, the metal-carbon bonds arise from covalent interaction between the π -electron cloud of unsaturated hydrocarbon and the vacant of filled d-orbitals which are present in transition metals.

Example - 15

Using valence bond approach, predict the shape and magnetism (i.e., paramagnetic or diamagnetic) of $[\text{Co}(\text{CO})_4]^+$. (Atomic number of Co = 27)

Sol. $[\text{Co}(\text{CO})_4]^+$

In its oxidation state of Co is +1.

Atomic number of Co = 27

Electronic configuration of Co atom :

$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$		$\uparrow\downarrow$		
3d					4s	4p	

Electronic configuration of Co (I) :

$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$		$\uparrow$		
3d					4s	4p	

sp^3 hybridized orbitals in Co (I) :

$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$	$\times$	$\times$	$\times$	$\times$
3d					s			

($\times$ = a lone pair of C of CO)

So, apparently the shape is tetrahedral and since the complex has two unpaired electrons, therefore, it is paramagnetic.

Example - 16

What is an 'ambident group'? Give an example.

Sol. It is the group which can combine at either ends. For example, $-\text{CN}$ group.

Example - 17

Write IUPAC name of the complex $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$.

Sol. Pentaquo chloro chromium (III) chloride.

Example - 18

Using the valence bond approach, predict the shape and magnetic character of $[\text{Fe}(\text{CN})_6]^{3-}$ ion. [At. no. of Fe = 26]

Sol.

	3d	4s	4p
Fe atom (Z = 26)	$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$	$\uparrow\downarrow$	
Ground state			
Fe_{3+} ion	$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$		
	d^2sp^3 hybridisation		
Hybridisation	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$		
$[\text{Fe}(\text{CN})_6]^{3-}$ ion	$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$	$\times\times$ $\times\times$ $\times\times$ $\times\times$ $\times\times$ $\times\times$	
	Six pairs of electrons from Six CN^- ions		

Fe^{3+} ion has configuration of $3d^5$. Two unpaired electrons paired up in 3d subshell. Two 3d-orbitals one 4s-orbital and three 4p-orbitals hybridise to give six equivalent d^2sp^3 hybrid orbitals. The complex ion has octahedral geometry and is paramagnetic due to the presence of one unpaired electron.

Example - 19

Name the following complex using IUPAC norms :

$[\text{Co}(\text{en})_2(\text{ONO})\text{Cl}]\text{Cl}$

Sol. Chloro bis (ethylene diamine) nitrilo cobalt (III) chloride.

Example - 20

Write a neutral molecule in which the central atom is sp^3d^2 hybridized.

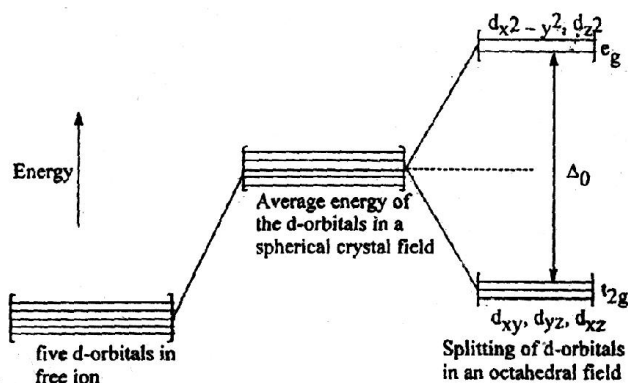
Sol. $\text{K}_3[\text{CoF}_6]$

Example - 21

Draw a figure to show splitting of degenerate d orbitals in an octahedral crystal field. How does the magnitude of Δ_o , decide the actual configuration of d orbitals in a complex entity?

Sol. Greater the value of Δ_o , the greater are the electron pairing. Figure shows the splitting of degenerate d orbitals in an

octahedral crystal field.



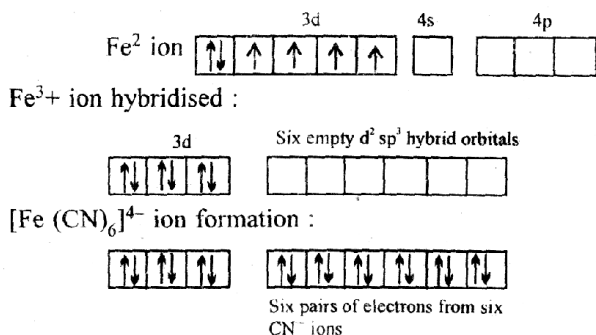
The actual configuration adopted is decided by the relative values of Δ_0 and P . Here P represents energy required for electron pairing in a single crystal.

- If $\Delta_0 < P$, we have the so called weak field, high spin situation. The fourth electron enters one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$. If now a fifth electron is added to a weak field coordination entity, the configuration becomes $t_{2g}^3 e_g^2$.
- If $\Delta_0 > P$, we have the strong field, low spin situation and pairing will occur in the t_{2g} level with the e_g level remaining unoccupied in entities of d^1 of d^6 ions.

Example - 22

- Give the IUPAC name of $[\text{PtCl}(\text{NH}_2\text{CH}_3)(\text{NH}_3)_2]\text{Cl}$.
- Compare the magnetic behaviour of the complex entities $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{FeF}_6]^{3-}$. ($\text{Fe} = 26$)

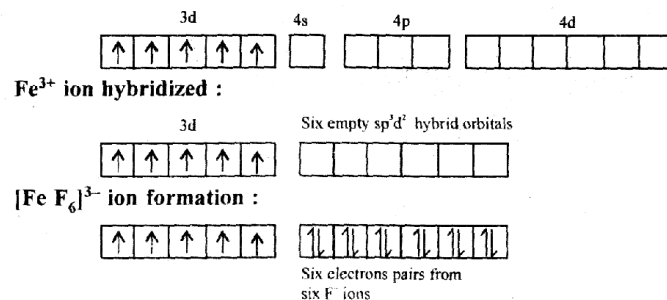
Sol. (a) Diamminechloro (methyl amine) platinum (II) chloride.
(b) $[\text{Fe}(\text{CN})_6]^{4-}$ ion



Since the complex ion does not contain any unpaired electron, so it is diamagnetic.

$[\text{FeF}_6]^{3-}$ ion

Fe³⁺ ion :



As the complex ion contains five unpaired electrons, it is highly paramagnetic in nature.

Example - 23

Write the IUPAC name of the linkage isomer of $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$.

Sol. IUPAC Name : Pentammine nitrito cobalt [III] chloride.

Example - 24

Write the formulas of the following coordination compounds :

- Tetraamine aquachloridocobalt (III) Chloride.
- Potassium tetrahydroxozincate (II).
- Tetracarbonylnickel (O).

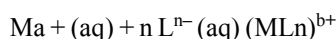
Sol. (i) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$
(ii) $\text{K}_2[\text{Zn}(\text{OH})_4]$
(iii) $[\text{Ni}(\text{CO})_4]$

Example - 25

Select a complex formation reaction and write an expression for the stability constant of the complex species. What information is conveyed regarding the strengths of ligands from the stability constant values of their complexes with a metal ion ? Illustrate your answer with examples of monodentate ligands.

Sol. Formation of complex : Please refer the formation of $[\text{Co}(\text{NH}_3)_3]^{3+}$, Oxd. number Co in this compound is +3.

For a general equation :



The equilibrium constant "K" is written as

$$K = \frac{[\text{MLn}]^{b+}}{[\text{M}^{a+}][\text{L}^{n-}]^n}$$

Example - 26

Write the IUPAC names of the following coordination compounds.

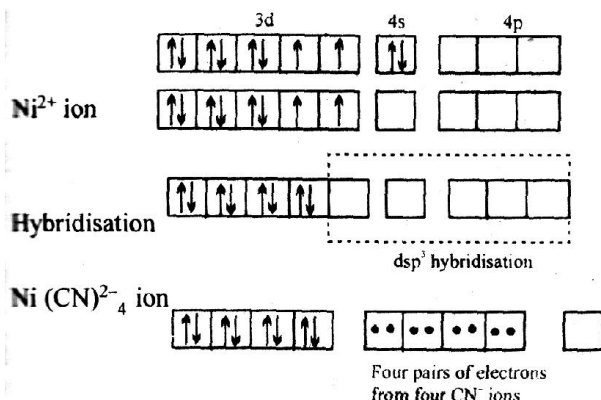
- (a) $[\text{CoCl}_2(\text{en})_2]\text{Cl}$
(b) $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$

- Sol.** (a) Dichloridobis (ethane -1, 2-diamine) Cobalt (III) Chloride.
(b) Pentaamine carbonatocobalt (III) Chloride.

Example - 27

Using valence bond theory of complexes, explain the geometry and magnetic nature of $[\text{Ni}^*(\text{CN})_4]^{2-}$, (At. No. Ni = 28)

- Sol.** Ni atom ($Z = 28$) Ground state

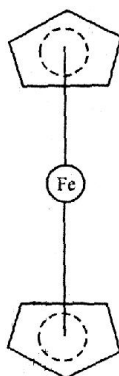


The Ni^{2+} ion is dsp^2 hybridised. Four pairs of electrons from four CN^- ions occupy the four vacant hybrid orbitals. So the complex ion has a square planar geometry. As it has no unpaired electron, so it is diamagnetic.

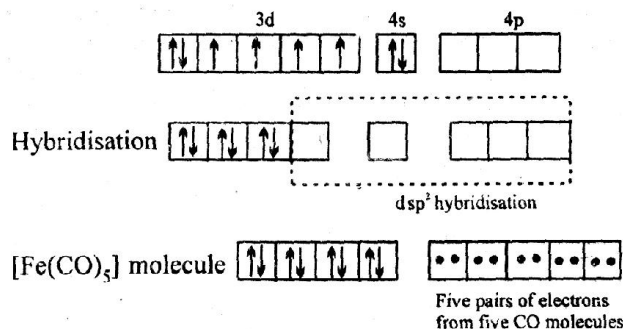
Example - 28

- (a) Draw the structure of ferrocene.
(b) Using valence bond theory explain the geometry and magnetic behaviour of pentacarbonyliron (0).

- Sol.** (a) Structure of ferrocene



- (b) Fe atom ($Z = 26$) Ground state

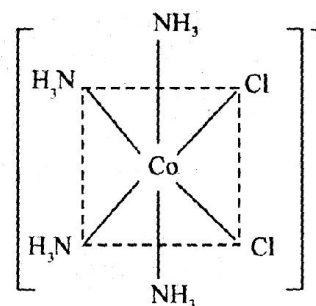


Fe atom is dsp^3 hybridised. Five pairs of electrons from five CO molecules occupy five hybrid orbitals. The $[\text{Fe}(\text{CO})_5]$ molecule has trigonal bipyramidal shape. As it has no unpaired electron, so it is diamagnetic.

Example - 29

- (a) What are ambidentate ligands? Give an example.
(b) Write IUPAC names of the following :
(i) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
(ii) $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$
(c) Draw the structure of cis isomer of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

- Sol.** (a) The ligands which can bond to the central metal atom/ion through two different atoms present in it are called ambidentate ligands. For example NO_2^- , which can attach itself through N atom or O atom.
(b) (i) Potassium trioxalatoferrate (III)
(ii) Hexaamine platinum (IV) chloride
(c) cis- $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$



Example - 30

Write the IUPAC names of the following coordination compounds :

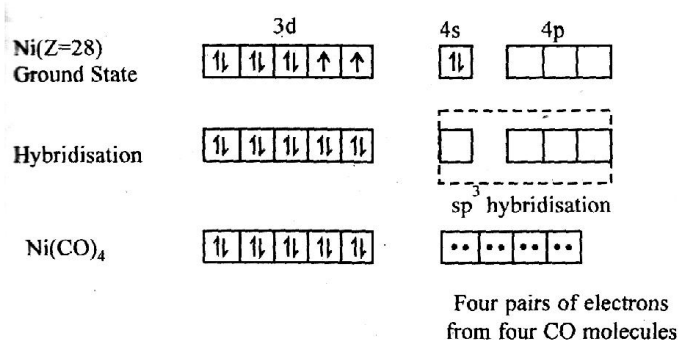
- $[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]_2(\text{SO}_4)_3$
- $[\text{Cr}(\text{NH}_3)_2(\text{H}_2\text{O})_3]\text{Cl}_3$
- $[\text{Ag}(\text{NH}_3)_2][\text{Ag}(\text{CN})_2]$

- Sol.** (a) Tris (ethane –1, 2-diamine) cobalt (III) sulphate
 (b) Triamminetriaquachromium (III) Chloride
 (c) Diamminesilver (I) dicyanargentate (I).

Example - 31

- Using valence bond theory, predict the shape and magnetic character of $[\text{Ni}(\text{CO})_4]$. $[\text{Ni} = 28]$
- Give one example of application of coordination compounds in medicine.

- Sol.** (a) Oxidation of NO in $\text{Ni}(\text{CO})_4$ is zero.



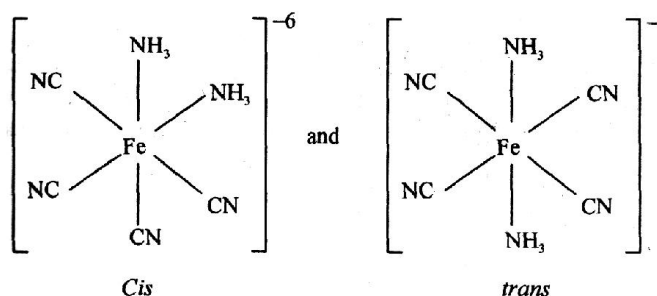
The complex molecule has tetrahedral shape because of sp³ hybridisation. As it has no unpaired electron, so it is diamagnetic.

- (b) The platinum complex $\text{cis-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ is known as Cisplatin and is used in the treatment of cancer.

Example - 32

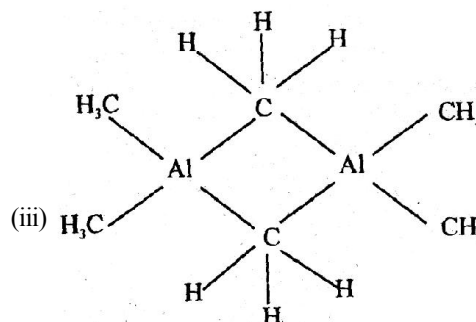
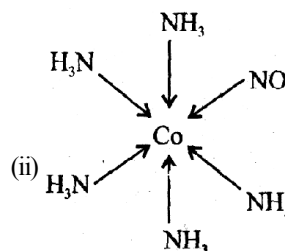
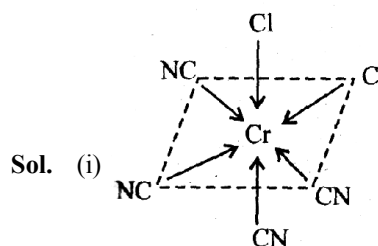
Draw structures of geometrical isomers of $[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^-$.

Sol.


Example - 33

Draw the structures of the following :

- Cis-dichlorotetracyanochromate (III)
- Pentaamminenitrito-N-cobalt (III)
- Hexamethyldialuminium

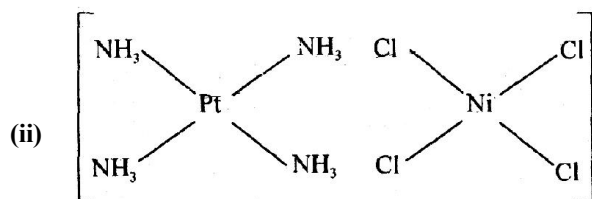
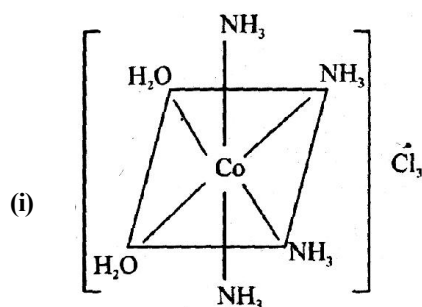

Example - 34

Write the name and draw the structures of each of the following complex compounds :

- $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_3$
- $[\text{Pt}(\text{NH}_3)_4][\text{NiCl}_4]$

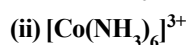
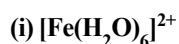
- Sol.** (i) Tetraaminediaqua cobalt (III) chloride
 (ii) Tetraamineplatinum (II) tetrachloronilcate (II).

Structures :

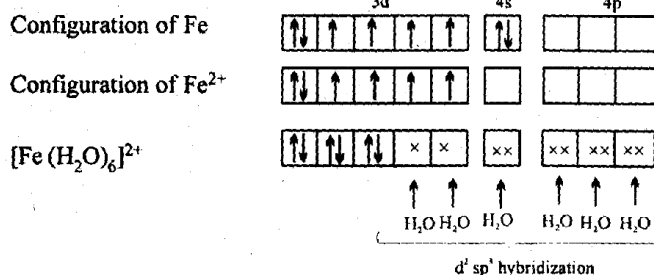
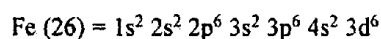
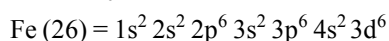


Example - 35

Describe for any two of the following complex ions, the type of hybridization, shape and magnetic property :

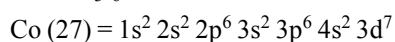
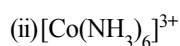


(At. Nos. Fe = 26, Co = 27, Ni = 28)

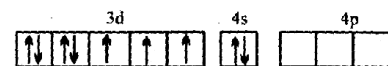
Sol. (i) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

Hybridization - $d^4 sp^3$

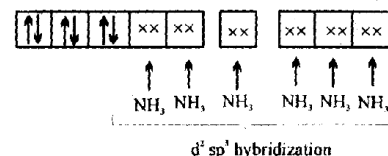
Shape - octahedral

Magnetic property - diamagnetic



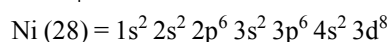
Configuration of Co


Configuration of Co^{2+}

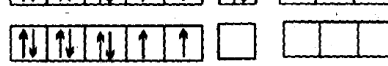
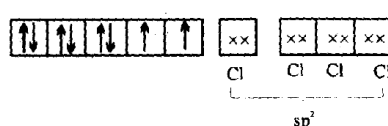
 $[\text{Co}(\text{NH}_3)_6]^{3+}$

Hybridization - $d^2 sp^3$

Shape - octahedral

Magnetic property - diamagnetic



Configuration of Ni


Configuration of Ni^{2+}

 $[\text{NiCl}_4]^{2-}$

Hybridization - sp^2

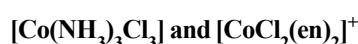
Shape - Tetrahedral

Magnetic property - paramagnetic

Example - 36

(a) What is the basis of formation of the spectro-chemical series ?

(b) Draw the structures of geometrical isomers of the following coordination complexes :

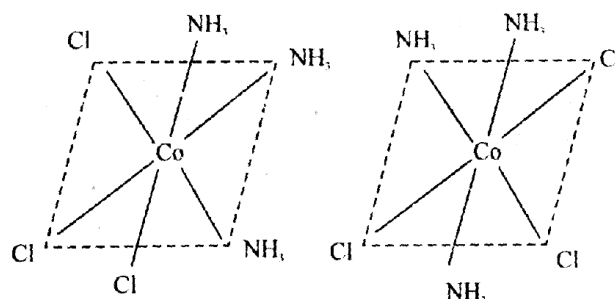
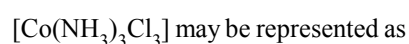


(en = ethylenediamine and atomic number of Co is 27)

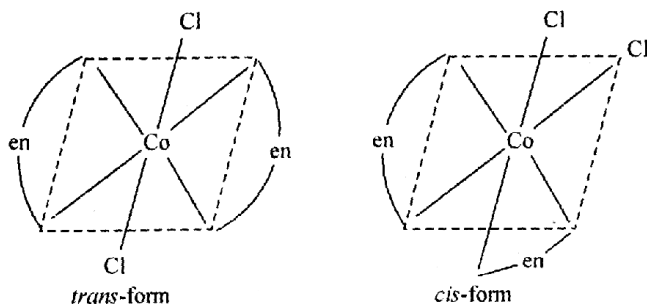
Sol. (a) Spectro-chemical series : The arrangement of ligands in order of their increasing field strength i.e. increasing crystal field splitting energy (CFSE) value is called spectro-chemical series.

Crystal field splitting energy is the basis of formation of the spectro-chemical series

(b) (i) The cis and trans isomers of



(ii) $[\text{CoCl}_2(\text{en})_2]^+$ ion exists in two geometrical isomers as shown below



Example - 37

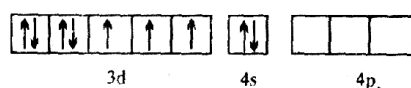
Using the valence bond approach, deduce the shape and magnetic character of

$[\text{Co}(\text{NH}_3)_6]^{3+}$ ion.

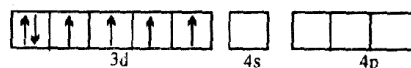
[Given : Atomic no. of Co = 27]

Sol. The electronic configuration of Co ($Z = 27$) is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$.

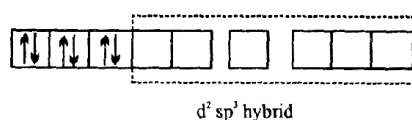
Co atom ($Z = 27$) Ground state



Orbitals of Co^{3+} ion



d^2sp^3 hybridised orbitals of Co^{3+}



$[\text{Co}(\text{NH}_3)_6]^{3+}$ (inner orbital or low spin complex)



Six pairs of electrons from six NH_3 molecules

$[\text{Co}(\text{NH}_3)_6]^{3+}$ complex has octahedral geometry and is diamagnetic because of the absence of unpaired electron.

Example - 38

Using the IUPAC norms, write the systematic name of the following :

$[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$

Sol. Hexaamine-cobalt (III) hexacyanochromate (III).

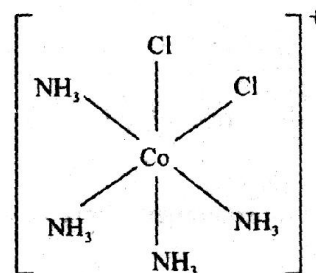
Example - 39

(a) Draw the structure and write the hybridisation state of Co in $\text{cis} - [\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

(b) Using the IUPAC norms name the following complex :

$[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{ONO})]\text{Cl}$

Sol. (a) Structure of $\text{cis} - [\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$



(b) Tetraamminechloridonitrito-N-cobalt (iii) Chloride.

Example - 40

Giving a suitable example for each, explain the following

(i) Crystal field splitting

(ii) Linkage isomerism

(iii) Ambidentate ligand

Sol. (i) **Crystal field splitting** : In a free transition metal ion, all the five d-orbitals are degenerate, but when it is involved in a complex formation, the degeneracy is split. This is called crystal field splitting.

(ii) **Linkage isomerism** : Linkage isomerism arises in a coordination compound containing ambidentate ligand. A simple example is provided by complexes containing the thiocyanate ligand, NCS^- , which may bind through the nitrogen to give $\text{M}-\text{NCS}$.

(iii) **Ambidentate ligand** : Unidentate ligands containing more than one coordinating atoms are called ambidentate ligands. For example,

NO_2^- , CN^- , SCN^- are all ambidentate ligands.

Example - 41

(a) Write the IUPAC name and indicate the shape of the complex ion $[\text{Co}(\text{en})_2\text{Cl}(\text{ONO})]^+$. [At. No. Co = 27]

(b) What is meant by chelate effect ?

Sol. (a) **IUPAC name** : Chlorobis (ethylenediamine) nitrito-O-cobalt (III) ion.

(b) **Chelate effect** : It may be noted that chelate ligands form more stable complexes than similar ordinary complexes in which the ligand acts as monodenate. This is called Chelate effect.

Example - 42

Give an example of linkage isomerism.

Sol. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$

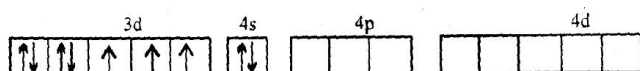
Example - 43

Using the Valence Bond theory predict the geometry and magnetic behaviour of $[\text{CoF}_6]^{3-}$. [At. No. of Co = 27]

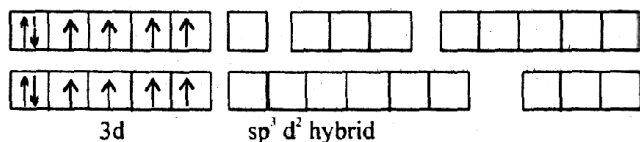
Sol. Formation of $[\text{CoF}_6]^{3-}$

Co (27) = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$

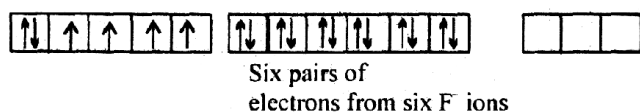
Configuration of Co



Co^{3+} ion sp^3d^2 hybridised orbitals of Co^{3+}



Formation of $[\text{CoF}_6]^{3-}$



Nature : Highly paramagnetic nature

Geometry : Octahedral

Example - 44

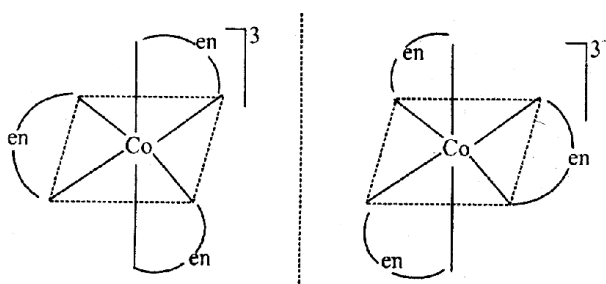
Write the structures and names of all the stereoisomers of the following compounds :

(i) $[\text{Co}(\text{en})_3]\text{Cl}_3$

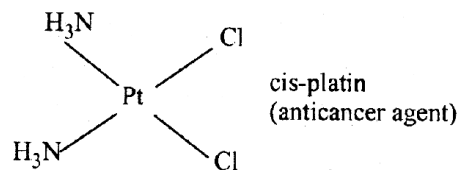
(ii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

(iii) $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

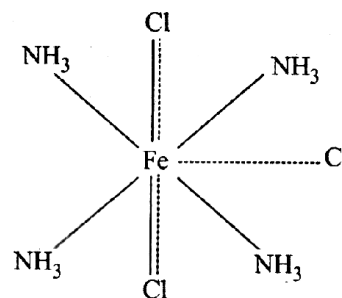
Sol. (i) $[\text{Co}(\text{en})_3]\text{Cl}_3$



(ii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$



(iii) $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$


Example - 45

Write the state of hybridization, the shape and the magnetic behaviour of the following complex entities :

(i) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

(ii) $[\text{Co}(\text{en})_3]\text{Cl}_3$

(iii) $\text{K}_2[\text{Ni}(\text{CN})_4]$

Sol. (i) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

IUPAC name : Pentaamminechlorido cobalt (III) Chloride

O.S. = +3, Co (III) : d^6

Stereochemistry : Octahedral

Conf ; t_{2g}^6

No. of unpaired electrons, $n = 0$

Magnetic moment = 0.

(ii) See

(iii) See

Example - 46

Write the IUPAC name of $[\text{Pt}(\text{NH}_3)_3(\text{NO})\text{Cl}_2]\text{Br}_2$.

Sol. Triamminedichloridonitrosoplatinum(II) bromide.

Example - 47

Write the IUPAC name of $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$.

Sol. Tetraamminedichloridoplatinum (IV) Dichloride (II).

Example - 48

Write the IUPAC name of $[\text{Cr}(\text{NH}_3)_5][\text{Co}(\text{CN})_6]$.

Sol. Hexa amido chromide (IV), Hexacyano Covalt (I).

Example - 49

Write the structure and write the hybridisation in

- (i) $[\text{Fe}(\text{CO})_5]$ (ii) $\text{trans}-[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
 (iii) $[\text{Ni}(\text{CO})_4]$ (iv) $\text{cis}-[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
 (v) $[\text{NiCl}_4]^{2-}$

- Sol.** (i) dsp^3 (trigonal bipyramidal)
 (ii) d^2sp^3 (octahedral)
 (iii) sp^3 (tetrahedral)
 (iv) d^2sp^3 (octahedral)
 (v) sp^3 (tetrahedral)

Example - 50

Write the formula of (i) hexaaquairon (II) sulphate (ii) potassium hexacyanoferrate (III) (iii) hexaammine platinum (IV) chloride (iv) potassium trioxalatoaluminate (III).

- Sol.** (i) $[\text{Fe}(\text{H}_2\text{O})_6]\text{SO}_4$
 (ii) $\text{K}_3[\text{Fe}(\text{CN})_6]$
 (iii) $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$
 (iv) $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$

Example - 51

Give names of two complexes which are used in medicines.

- Sol.** (i) EDTA, used in the treatment of lead poisoning.
 (ii) cis-platin, $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, used in the treatment of cancer.

Example - 52

Write down the heterogeneous catalyst involved in the polymerisation of ethylene.

Sol. Zeigler-Natta catalyst (Trialkyl aluminium + Titanium tetrachloride). Ethylene is passed under pressure into a hydrocarbon solvent containing a suspension of $(\text{C}_2\text{H}_5)_3\text{Al} + \text{TiCl}_4$ at about 70°C .

Example - 53

Give an example of linkage isomerism.

Sol. $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$

Example - 54

Give an example of coordination isomerism.

Sol. $[\text{Cr}(\text{NH}_3)_6][\text{CoF}_6]$ and $[\text{Co}(\text{NH}_3)_6][\text{CrF}_6]$

Example - 55

Give an example of ionization isomerism.

Sol. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Double Salts

- Which of the following is a coordination compound?
(a) $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (b) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
(c) $\text{NaNH}_4\text{HPO}_4 \cdot 4\text{H}_2\text{O}$ (d) $\text{K}_3\text{Fe}(\text{CN})_6 \cdot 2\text{H}_2\text{O}$
- Which of the following is a double salt?
(a) Alum (b) Chrome alum
(c) Microcosmic salt (d) All of these
- When potash alum is dissolved in water, the total number of ions produced is
(a) four (b) eight
(c) ten (d) thirty-two
- Which of the following statements is correct with regard to a complex ion?
(a) A complex ion consists of a central ion bonded to two or more donor ions or molecules, usually does not dissociate into simple ions or molecules even in a solution and exhibits properties different from its constituent ions or molecules.
(b) The donor ions and molecules which coordinate with the central atom or ion in a complex are called ligands.
(c) The sum of the number of electrons present in the central metal ion or atom and those donated by the ligands is called the effective atomic number of the central metal atom and this number is usually the same as the atomic number of the next higher noble gas.
(d) All of these
- The neutral molecules or ions which coordinate with the central atom or ion in a complex ion are called
(a) combinants (b) cryptands
(c) spherands (d) ligands
- The number of ions formed from a molecule of potassium ferrocyanide in aqueous solution is
(a) 2 (b) 3
(c) 5 (d) 1

Oxidation State

- The oxidation number of Fe in $\text{K}_3[\text{Fe}(\text{CN})_5\text{NO}]$ is
(a) +4 (b) +2
(c) +3 (d) +6

- The oxidation number of Co in $[\text{Co}(\text{en})_3](\text{SO}_4)_3$ is
(a) +2 (b) +4
(c) +3 (d) +6
- The oxidation number of Cr in $[\text{Cr}(\text{C}_6\text{H}_6)_2]$ is
(a) +2 (b) +6
(c) +3 (d) zero
- The coordination number of Ag in $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ is
(a) one (b) two
(c) three (d) zero
- The coordination number of Pt in $\text{Pt}(\text{NH}_3)_4\text{Cl}_2$ is
(a) 12 (b) 2
(c) 6 (d) 4
- The coordination number of Fe in $\text{K}_4[\text{Fe}(\text{CN})_6]$ is
(a) four (b) three
(c) six (d) ten
- In the complex ion $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$
(a) Fe is in the +1 oxidation state, and NO coordinates as NO^+ (nitrosonium ion)
(b) Fe is in the +2 oxidation state, and NO coordinates as neutral NO (nitrosyl) radical
(c) Fe is in the +3 oxidation state and NO coordinates as NO^-
(d) Fe is in the +2 oxidation state and NO coordinates as NO^+

Nomenclature

- The IUPAC name for the complex compound $\text{Li}[\text{AlH}_4]$ is
(a) lithium aluminium hydride
(b) hydrido-aluminiumlithium (III)
(c) lithium tetrahydridoaluminate(III)
(d) lithium tetrahydridoaluminate(I)
- The IUPAC name for the coordination compound $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$ is
(a) sodium silverthiosulphate (I)
(b) sodium silverrhyposulphate
(c) sodium bis[argentothiosulphate(I)]
(d) sodium bis (thiosulphato) argentate (I)

16. The IUPAC name for the coordination compound $\text{Fe}(\text{C}_5\text{H}_5)_2$ is
 (a) ferrocene
 (b) bis(cyclopentadienyl)iron (II)
 (c) dicyclopentadienyliron (0)
 (d) bis(cyclopentyl)iron (II)
17. The IUPAC name for the coordination compound $[\text{CuCl}_2(\text{CH}_3\text{NH}_2)_2]$ is
 (a) dimethylamine copper (II) chloride
 (b) bis(dimethylamine) copper (II) chloride
 (c) dichlorobis (methylamine) copper (II)
 (d) dichlorobis(dimethylamine) copper (II)
18. The IUPAC name for $[\text{Al}(\text{OH})(\text{H}_2\text{O})_5]^{2+}$ is
 (a) pentahydroaluminium hydroxide
 (b) aquometaaluminate ion
 (c) pentaquaaluminate(III) hydroxide
 (d) pentaquahydroxoaluminium (III)
19. The IUPAC name for $[\text{Ru}(\text{NH}_3)_5\text{N}_2]\text{Cl}_2$ is
 (a) pentaammine(dinitrogen)ruthenium (II) chloride
 (b) pentaammine (nitrogen)ruthenium (II) chloride
 (c) rutheniumpentaammine(II) dichlorodinitrogen
 (d) pentaammine (azido) ruthenium (II) dichloride
20. The IUPAC name for $\text{K}[\text{SbCl}_5\text{C}_6\text{H}_5]$ is
 (a) potassium phenylchloroantimonate (V)
 (b) potassium benzalantimony (II) chloride
 (c) potassium pentachloro(phenyl) antimonate (V)
 (d) luto antimonate (V)
21. The IUPAC name for $[\text{Pt}(\text{NH}_3)_3(\text{Br})(\text{NO}_2)(\text{Cl})]\text{Cl}$ is
 (a) triamminechlorobromonitroplatinum(IV) chloride
 (b) triamminebromochloronitroplatinum(IV) chloride
 (c) triamminenitrochlorobromoplatinum(IV) chloride
 (d) triamminechloronitrobromoplatinum(IV) chloride
22. The IUPAC name of $\text{Ba}[\text{BrF}_4]_2$ is
 (a) barium bromofluoride (III)
 (b) barium bis[tetrafluorobrominate(III)]
 (c) barium bis(tetrafluorobromate)(V)
 (d) barium tetrafluorobromate(III)
23. The IUPAC name for $\text{K}_2[\text{Cr}(\text{CN})_2\text{O}_2(\text{O}_2)\text{NH}_3]$ is
 (a) potassium amminedicyanodioxoperoxochromate (VI)
 (b) potassium amminedicyanotetroxochromium (III)
 (c) potassium amminedicyanochromate (IV)
 (d) potassium aminocyanodiperoxochromate (VI)
24. The IUPAC name for $\text{K}_2[\text{OsCl}_5\text{N}]$ is
 (a) potassium pentachloroazidoosmate (VIII)
 (b) potassium pentachloroazoosmate(VI)
 (c) potassium pentachloronitridoosmate (VI)
 (d) potassium nitroosmate (III)
- Ligand**
25. Which of the following is a bidentate ligand?
 (a) Oxalato (b) Carbonate
 (c) Glycinato (d) All of these
26. Which of the following is a hexadentate ligand?
 (a) Acetylacetonato
 (b) 8-Hydroxy quinolinato
 (c) Ethylenediamine tetraacetato
 (d) Ethylenediamine
- EAN**
27. Which of the following complex ions obeys Sidgwick's effective atomic number (EAN) rule?
 (a) $[\text{Fe}(\text{CN})_6]^{3-}$ (b) $[\text{Fe}(\text{CN})_6]^{4-}$
 (c) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (d) $[\text{Ni}(\text{en})_3]^{2+}$
28. Which of the following complex ions violates the EAN rule?
 (a) $[\text{Pt}(\text{NH}_3)_6]^{4+}$ (b) $[\text{Fe}(\text{CO})_5]$
 (c) $[\text{Fe}(\text{CN})_6]^{4-}$ (d) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
29. The effective atomic number for the complex ion $[\text{Ni}(\text{CN})_4]^{2-}$ is
 (a) 34 (b) 38
 (c) 36 (d) 54
30. The effective atomic number for the complex ion $\text{Pd}(\text{NH}_3)_6^{4+}$ is
 (a) 50 (b) 86
 (c) 54 (d) 36

VB

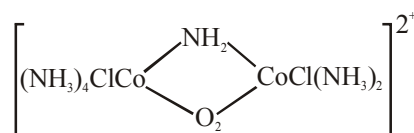
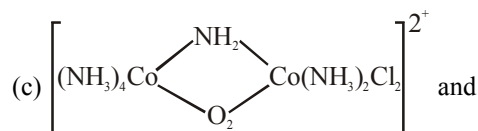
31. The structures of Ni(CO)_4 and $\text{Ni(PPh}_3)_2\text{Cl}_2$ are
(a) square planar
(b) tetrahedral and square planar respectively
(c) tetrahedral
(d) square planar and tetrahedral respectively
32. The hybridization states of the central atom in the complexes Fe(CN)_6^{3-} , Fe(CN)_6^{4-} and $\text{Co(NO}_2)_6^{3-}$ are
(a) d^2sp^3 , sp^3d^2 and dsp^2 respectively
(b) d^2sp^3 , sp^3d and sp^3d^2 respectively
(c) d^2sp^3 , sp^3 and d^4s^2 respectively
(d) all d^2sp^3
33. Which of the following complex ions is paramagnetic ?
(a) $[\text{Fe(CN)}_6]^{4-}$ (b) $[\text{Fe(CN)}_6]^{3-}$
(c) $[\text{Fe(CN)}_5\text{NO}]^{2-}$ (d) $[\text{Co(NO}_2)_6]^{3-}$
34. The hybridization states of the central atom in the complex ions $[\text{FeF}_6]^{3-}$, $[\text{Fe(H}_2\text{O)}_6]^{3+}$ and $[\text{Ni(NH}_3)_6]^{2+}$ are
(a) sp^3d^2 , dsp^2 and d^4s^2 respectively
(b) all $3d^24s4p^3$
(c) all $4s4p^34d^2$
(d) sp^3d^2 , dsp^3 and p^4d^2 respectively
35. The formation of the complex ion $[\text{Co(NH}_3)_6]^{3+}$ involves sp^3d^2 hybridization of Co^{3+} . Hence, the complex ion should possess
(a) octahedral geometry
(b) tetrahedral geometry
(c) square planar geometry
(d) tetragonal geometry
36. Which of the following complex ions is diamagnetic?
(a) $[\text{FeF}_6]^{3-}$ (b) $[\text{Cu(NH}_3)_6]^{2+}$
(c) $[\text{Cr(H}_2\text{O)}_6]^{2+}$ (d) $[\text{Zn(NH}_3)_6]^{2+}$
37. Which of the following octahedral complex ions is diamagnetic ?
(a) $[\text{Cr(CO)}_6]$ (b) $[\text{Mn(CN)}_6]^{4-}$
(c) $[\text{Ni(NH}_3)_6]^{2+}$ (d) $[\text{CoF}_6]^{3-}$
38. Which of the following statements is correct ?
(a) The $[\text{Ni(CN)}_4]^{2-}$ ion has a tetrahedral geometry and is diamagnetic.
(b) The $[\text{Ni(CN)}_4]^{2-}$ ion has a square planar geometry and is paramagnetic.
(c) The $[\text{Ni(CN)}_4]^{2-}$ ion has a square planar geometry and is diamagnetic.
(d) The $[\text{Cu(NH}_3)_4]^{2+}$ ion has a tetrahedral geometry and is diamagnetic.
39. Which of the following is diamagnetic?
(a) $[\text{Ni(NH}_3)_4]^{2+}$ (b) $[\text{NiCl}_4]^{2-}$
(c) $[\text{Ni(H}_2\text{O)}_4]^{2+}$ (d) $[\text{Ni(CO)}_4]$
40. Which of the following complex ions/molecules of nickel is paramagnetic?
(a) $[\text{Ni(NH}_3)_4]^{2+}$ (b) $\text{Ni(dimethylglyoxime)}_2$
(c) $[\text{Ni(CO)}_4]$ (d) None of these
41. Which of the following statements is incorrect?
(a) Most of the four-coordinated complexes of Ni^{2+} ion are square planar rather than tetrahedral.
(b) The $[\text{Fe(H}_2\text{O)}_6]^{3+}$ ion is more paramagnetic than the $[\text{Fe(CN)}_6]^{3-}$ ion.
(c) Square planar complexes are more stable than octahedral complexes.
(d) The $[\text{Fe(CN)}_6]^{4-}$ ion is paramagnetic but $[\text{Fe(CN)}_6]^{3-}$ is diamagnetic.
42. The compound which does not show paramagnetism is
(a) $[\text{Cu(NH}_3)_4]\text{Cl}_2$ (b) $[\text{Ag(NH}_3)_2]\text{Cl}$
(c) NO (d) NO_2
43. Among TiF_6^{2-} , CoF_6^{3-} , Cu_2Cl_2 and NiCl_4^{2-} (atomic numbers of Ti = 22, Co = 27, Cu = 29, Ni = 28), the colourless species are
(a) CoF_6^{3-} and NiCl_4^{2-} (b) TiF_6^{2-} and CoF_6^{3-}
(c) Cu_2Cl_2 and NiCl_4^{2-} (d) TiF_6^{2-} and Cu_2Cl_2
44. Among $[\text{Ni(CO)}_4]$, $[\text{Ni(CN)}_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$
(a) $[\text{Ni(CO)}_4]$ and $[\text{Ni(CN)}_4]^{2-}$ are diamagnetic and $[\text{Ni(CN)}_4]^{2-}$ is paramagnetic
(b) $[\text{NiCl}_4]^{2-}$ and $[\text{Ni(CN)}_4]^{2-}$ are diamagnetic and $[\text{Ni(CO)}_4]$ is paramagnetic
(c) $[\text{Ni(CO)}_4]$ and $[\text{Ni(CN)}_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic
(d) $[\text{Ni(CO)}_4]$ is diamagnetic and $[\text{NiCl}_4]^{2-}$ and $[\text{Ni(CN)}_4]^{2-}$ are paramagnetic.

Organometallic

45. Zeise salt, an organometallic compound, has the formula
- (a) $(C_6H_6)_2Cr^+ AlCl_4^-$ (b) $(CH_3)_2AlF$
(c) $Ni(CO)_4$ (d) $K^+[PtC_2H_4Cl_3]H_2O$
46. Which of the following mixtures is known as Ziegler–Natta catalyst ?
- (a) $Al(OCH_3)_3 + TiCl_4$ (b) $(C_2H_5)_3Al + TiCl_4$
(c) $\{(CH_3)_2CHO\}_3Al + TiCl_4$
(d) $LiCH_3 + TiCl_4$
47. Which of the following is an organometallic compound?
- (a) Cyclobutadiene
(b) Thiotetraamminecopper (II) sulphate
(c) Potassium tetrafluorooxochromate (V)
(d) Bis (cyclopentadienyl) iron (II)
48. Which of the following is not an organometallic compound?
- (a) Ferrocene (b) Ruthenocene
(c) Beryllium acetylacetonate
(d) bis(benzene) chromium
49. Which is not a π -bonded complex?
- (a) Zeise salt (b) Ferrocene
(c) bis(benzene) chromium (d) Tetraethyl lead
50. Which of the following is an organometallic compound?
- (a) Lithium methoxide (b) Lithium acetate
(c) Lithium dimethylamide (d) Methylolithium
51. Which of the following statements is correct for ferrocene?
- (a) It has a sandwich structure.
(b) The iron atom is placed between the two cyclopentadiene rings.
(c) It has a pentagonal antiprismatic arrangement.
(d) All of these
52. Which of the following ligands can form an organometallic compound?
- (a) NO (b) C_2H_4
(c) SCN^- (d) Cl^-
53. The organometallic compound formed during the purification of Ni by the Mond process is
- (a) $K_4[Ni(CN)_4]$ (b) $[Ni(en)_3]^{2+}$
(c) $Ni(CO)_4$ (d) $NiBr_3(PMe_3)_3$

Structural Isomerism

54. Which one of the following coordination compounds exhibits ionization isomerism?
- (a) $[Cr(NH_3)_6]Cl_3$ (b) $[Cr(en)_3]Cl_3$
(c) $[Cr(en)_3]Cl_3$ (d) $[Co(NH_3)_5Br]SO_4$
55. The pair $[Co(NH_3)_5NO_3]SO_4$ and $[Co(NH_3)_5SO_4]NO_3$ will exhibit
- (a) hydrate isomerism (b) linkage isomerism
(c) ionization isomerism (d) coordinate isomerism
56. What type of isomerism could you assign to the following pair of complex compounds?
 $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$
- (a) coordination isomerism
(b) Polymerization isomerism
(c) Coordination position isomerism
(d) Ligand isomerism
57. Which one among the following pairs of complex compounds indicates linkage isomerism?
- (a) $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$
(b) $[Cu(NH_3)_4][PtCl_4]$ and $[CuCl_4][Pt(NH_3)_4]$



- (d) $[Co(NO_2)(NH_3)_5]Cl_2$ and $[Co(ONO)(NH_3)_5]Cl_2$
58. How many hydrate isomers are possible with the formula $CrCl_3 \cdot 6H_2O$?
- (a) Six (b) Four
(c) Three (d) Two
59. A coordination compound of cobalt has the molecular formula containing five ammonia molecules, one nitro group and two chlorine atoms for one cobalt atom. One mole of this compound produces three moles of ions in an aqueous solution. The aqueous solution on treatment with an excess of $AgNO_3$ gives two moles of $AgCl$ as a precipitate. The formula of this complex would be
- (a) $[Co(NH_3)_4NO_2Cl][NH_3Cl]$
(b) $[Co(NH_3)Cl][ClONO_2]$
(c) $[Co(NH_3)_5NO_2]Cl_2$ (d) $[Co(NH_3)_5][Cl(NO_2)_2]$

60. The compounds $[\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_3]$, $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$, H_2O and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ exhibit

- (a) linkage isomerism (b) geometrical isomerism
(c) ionization isomerism (d) hydrate isomerism

Stereo Isomerism

61. Which of the following will have three stereoisomeric forms?



(where en = ethylene diamine)

- (a) (iv) and (iii) (b) (iv) and (i)
(c) (iii) and (ii) (d) (i) and (ii)

62. Which of the following complex compounds exhibits cis-trans isomerism?

- (a) $[\text{PtCl}_2(\text{NH}_3)_2]$ (b) $[\text{PdCl}_2\text{BrI}]$
(c) $[\text{Pt}(\text{NH}_3)(\text{py})(\text{Cl})(\text{Br})]$ (d) All of these

63. Which of the following complex compounds exhibits *cis-trans* isomerism?

- (a) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]$ (b) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
(b) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (d) All of these

64. Which of the following types of square planar complexes can show geometrical isomerism (where M = metal, a, b = achiral ligands)?

- (a) Ma_4 (b) Ma_3b
(c) Ma_2b_2 (d) Mab_3

65. Tetrahedral complexes of the types of $[\text{Ma}_4]$ and $[\text{Ma}_3\text{b}]$ (where M = metal, a, b = achiral ligands) are not able to show optical isomerism because

- (a) these molecules/ions possess C_n axis of symmetry
(b) these molecules/ions possess a plane of symmetry and hence are achiral
(c) these molecules possess a centre of symmetry
(d) these molecules/ions have nonsuperimposable mirror images

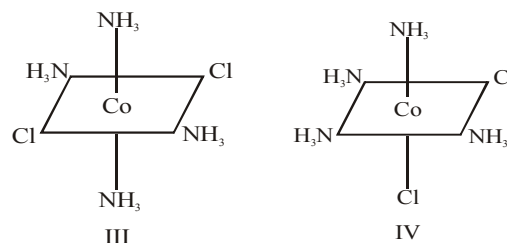
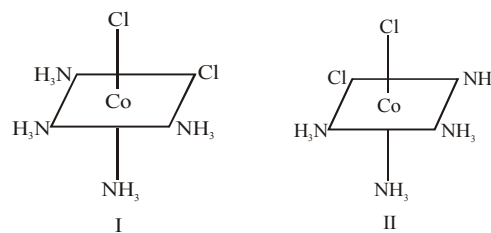
66. How many geometrical isomers are possible for the square planar complex $[\text{Pt}(\text{NO}_2)(\text{py})(\text{NH}_3)(\text{NH}_2\text{OH})]\text{NO}_2$?

- (a) Four (b) Five
(c) Eight (d) Three

67. Which of the following types of octahedral complexes will exhibit geometrical isomerism (where M = metal, a, b = achiral ligands)?

- (a) $[\text{Ma}_6]$ (b) $[\text{Ma}_5\text{b}]$
(c) $[\text{M}(\text{aa})_3]$ (d) $[\text{Ma}_4\text{b}_2]$

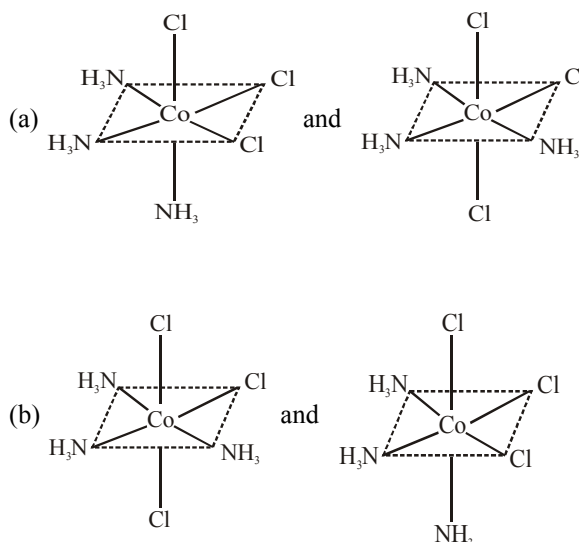
68. consider the following spatial arrangements of the octahedral complex ion $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

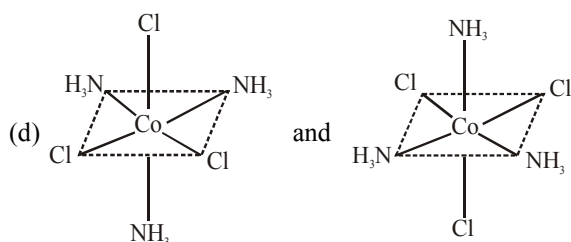
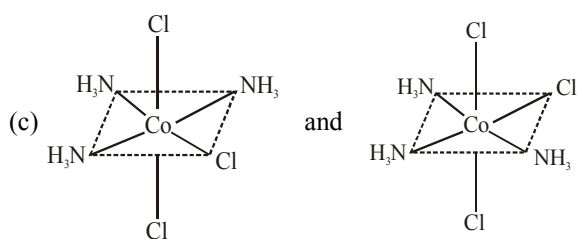


Which of the following statements is incorrect regarding these structures?

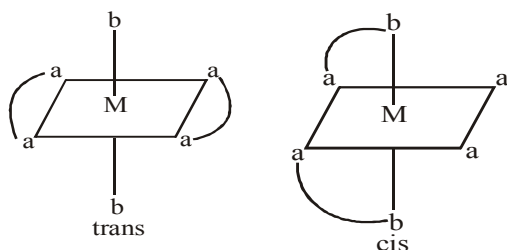
- (a) I and II are enantiomers.
(b) II and III are cis and trans isomers respectively.
(c) III and IV are trans and cis isomers respectively.
(d) II and IV have identical structures.

69. Which of the following pairs of structures represent facial and meridional isomers (geometrical isomers) respectively?





70. Which of the following statements is correct regarding the chirality (optical isomerism) of the cis and trans isomers of the type $M(aa)_2b_2$ (M stands for a metal, a and b are achiral ligands and aa is a bidentate ligand)?



- (a) The trans form is achiral and optically inactive while the cis form is chiral and exists in two enantiomeric forms.
- (b) Both the cis and trans forms are achiral and optically inactive.
- (c) The trans form is chiral and exists in two enantiomeric forms while the cis form is achiral and optically inactive.
- (d) Both the cis and trans forms are chiral, and each exists in two enantiomeric forms.
71. The tris (ethylenediamine) cobalt (III) cation, $[Co(en)_3]^{3+}$, can have
- (a) three stereoisomers, all chiral and optically active
- (b) two chiral stereoisomers (enantiomers)
- (c) three stereoisomers, all achiral
- (d) two stereoisomers, both achiral

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

- Which of the following ions has the maximum magnetic moment ? (2002)
(a) Mn^{2+} (b) Fe^{2+}
(c) Ti^{2+} (d) Cr^{2+}
- A square planar complex is formed by hybridisation of the following atomic orbitals (2002)
(a) s, p_x, p_y, p_z (b) s, p_x, p_y, p_z, d
(c) d, s, p_x, p_y (d) s, p_x, p_y, p_z, d, d
- Type of isomerism shown by $[\text{Cr}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ is (2002)
(a) optical (b) ionisation
(c) geometrical (d) linkage
- Most stable ion is (2003)
(a) $[\text{Fe}(\text{OH})_6]^{3-}$ (b) $[\text{Fe}(\text{OH})_6]^{3-}$
(c) $[\text{Fe}(\text{CN})_6]^{3-}$ (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
- In the coordination compound, $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is (2003)
(a) +2 (b) 0
(c) -1 (d) +1
- One mole of the complex Compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of AgCl (s). The structure of the complex is (2003)
(a) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2\cdot\text{NH}_3$
(b) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
(c) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]\cdot 2\text{NH}_3$
(d) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}\cdot\text{NH}_3$
- Ammonia forms complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in alkaline solution but not in acidic solution. What is the reason for it ? (2003)
(a) Copper hydroxide is an amphoteric substance
(b) In acidic solution hydration protects copper ions
(c) In acidic solution protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
(d) In alkaline solution insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
- Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect ? (2004)
(a) Chlorophylls are green pigments in plants and contain calcium
(b) Haemoglobin is the red pigment of blood and contains iron
(c) Cyanocobalamin is B_{12} and contains cobalt
(d) Carboxypeptidase-A is an enzyme and contains zinc
- Which one of the following complexes is an outer orbital complex ?
(Atomic numbers : $\text{Mn} = 25, \text{Fe} = 26, \text{Co} = 27, \text{Ni} = 28$) (2004)
(a) $[\text{Fe}(\text{CN})_6]^{4-}$ (b) $[\text{Mn}(\text{CN})_6]^{4-}$
(c) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (d) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
- The coordination number of central metal atom in a complex is determined by (2004)
(a) the number of ligands around a metal ion bonded by sigma bonds
(b) the number of ligands around a metal ion bonded by pi bonds
(c) the number of ligands around a metal ion bonded by sigma and pi bonds both
(d) the number of only anionic ligands bonded to the metal ion.
- Among the properties, (A) reducing, (B) oxidizing, (C) complexing, the set of properties shown by CN^- ion towards metal species is (2004)
(a) A, B (b) B, C
(c) C, A (d) A, B, C
- The correct order of magnetic moments (spin only values in B.M.) is (2004)
(a) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > \text{Fe}(\text{CN})_6]^{4-}$
(b) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
(c) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$
(d) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$

13. Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them ? (2004)
- (a) $(n-1)d^8ns^2$ (b) $(n-1)d^5ns^1$
(c) $(n-1)d^3ns^2$ (d) $(n-1)d^5ns^2$
14. Which one of the following has largest number of isomers ? (2004)
- (R = alkyl group, en = ethylenediamine)
- (a) $[\text{Ru}(\text{NH}_3)_4\text{Cl}_2]^+$ (b) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$
(c) $[\text{Ir}(\text{PR}_3)_2\text{H}(\text{CO})]^{2+}$ (d) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
15. Which of the following compounds shows optical isomerism ? (2005)
- (a) $[\text{Co}(\text{CN})_6]^{3-}$ (b) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$
(c) $[\text{ZnCl}_4]^{2-}$ (d) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
16. Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour? (2005)
- (At. no. Cr = 24, Mn = 25, Fe = 26, Co = 27)
- (a) $[\text{Co}(\text{CN})_6]^{3-}$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$
(c) $[\text{Mn}(\text{CN})_6]^{3-}$ (d) $[\text{Cr}(\text{CN})_6]^{3-}$
17. The value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM. The correct one is (2005)
- (a) d^5 (in strong ligand field)
(b) d^3 (in weak as well as in strong fields)
(c) d^4 (in weak ligand field)
(d) d^4 (in strong ligand field)
18. The oxidation state of Cr in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ is (2005)
- (a) +1 (b) 0
(c) +3 (d) +2
19. The IUPAC name of the coordination compound $\text{K}_3[\text{Fe}(\text{CN})_6]$ is (2005)
- (a) potassium hexacyanoiron (II)
(b) tripotassium hexacyanoiron (II)
(c) potassium hexacyanoferrate (II)
(d) potassium hexacyanoferrate (III)
20. Nickel ($Z = 28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron (s) in the nickel and geometry of this complex ion are respectively (2006)
- (a) two, tetrahedral (b) one, square planar
(c) two, square planar (d) one, tetrahedral
21. The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is (2006)
- (a) nitrito-N-pentaamminecobalt (II) chloride
(b) pentaamminenitrito-N-cobalt (II) chloride
(c) pentaamminenitrito-N-cobalt (III) chloride
(d) nitrito-N-pentaamminecobalt (III) chloride.
22. In $\text{Fe}(\text{CO})_5$, the Fe – C bond possesses [2006]
- (a) π -character only (b) both σ and π -characters
(c) ionic character (d) σ -character only
23. Which one of the following has a square planar geometry ? (2007)
- (At. no. Co = 27, Ni = 28, Fe = 26, Pt = 78)
- (a) $[\text{CoCl}_4]^{2-}$ (b) $[\text{FeCl}_4]^{2-}$
(c) $[\text{NiCl}_4]^{2-}$ (d) $[\text{PtCl}_4]^{2-}$
24. In which of the following octahedral complexes of Co (at no 27), will the magnitude of Δ_o be the highest? (2008)
- (a) $[\text{Co}(\text{CN})_6]^{3-}$ (b) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
(c) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (d) $[\text{Co}(\text{NH}_3)_6]^{3+}$
25. The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4)]\text{NO}_2$ (where (en) is ethylene diamine) are, respectively. (2008)
- (a) 6 and 2 (b) 4 and 2
(c) 4 and 3 (d) 6 and 3
26. Which of the following pairs represents linkage isomers ? (2009)
- (a) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$
(b) $[\text{Pd}(\text{PPh}_3)_2(\text{NCS})_2]$ and $[\text{Pd}(\text{PPh}_3)_2(\text{SCN})_2]$
(c) $[\text{CO}(\text{NH}_3)_5]\text{NO}_3\text{SO}_4$ and $[\text{CO}(\text{NH}_3)_5\text{SO}_4]\text{NO}_3$
(d) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$ and $[\text{PtBr}_2(\text{NH}_3)_4]\text{Cl}_2$

27. In context with the transition elements, which of the following statements is incorrect ? (2009)
- (a) In addition to the normal oxidation state, the zero oxidation state is also shown by these elements in complexes.
- (b) In the highest oxidation states, the transition metal shows basic character and form cationic complexes.
- (c) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4s and 3d electrons are used for bonding.
- (d) Once the d^5 configuration is exceeded, the tendency to involve all the 3d electrons in bonding decreases.
28. Which of the following has an optical isomer ? (2009)
- (a) $[\text{Co}(\text{NH}_3)_3\text{Cl}]^+$ (b) $[\text{Co}(\text{en})(\text{NH}_3)_2]^{2+}$
- (c) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ (d) $[\text{Co}(\text{en})_2(\text{NH}_3)_2]^{3+}$
29. Which one of the following has an optical isomer ? (en = ethylenediamine) (2010)
- (a) $[\text{Zn}(\text{en})(\text{NH}_3)_2]^{2+}$ (b) $[\text{Co}(\text{en})_3]^{3+}$
- (c) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ (d) $[\text{Zn}(\text{en})_2]^{2+}$
30. Which one of the following complex ions has geometrical isomers ? (2011)
- (a) $[\text{Co}(\text{en})_3]^{3+}$ (b) $[\text{Ni}(\text{NH}_3)_5\text{Br}]^+$
- (c) $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$ (d) $[\text{Cr}(\text{NH}_3)_4(\text{en})]^{3+}$
31. Among the ligands NH_3 , en, CN^- and CO, the correct order of their increasing field strength, is (2011)
- (a) $\text{CO} < \text{NH}_3 < \text{en} < \text{CN}^-$ (b) $\text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
- (c) $\text{CN}^- < \text{NH}_3 < \text{CO} < \text{en}$ (d) $\text{en} < \text{CN}^- < \text{NH}_3 < \text{CO}$
32. The magnetic moment (spin only) of $[\text{NiCl}_4]^{2-}$ is (2011)
- (a) 1.82 BM (b) 5.46 BM
- (c) 2.82 BM (d) 1.41 BM
33. Which of the following facts about the complex $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is wrong ? (2011)
- (a) The complex involves d^2sp^3 hybridization and is octahedral in shape.
- (b) The complex is paramagnetic.
- (c) The complex is an outer orbital complex.
- (d) The complex gives white precipitate with silver nitrate solution.
34. Which among the following will be named as dibromidobis-(ethylenediamine) chromium (III) bromide ? (2012)
- (a) $[\text{Cr}(\text{en})_3]\text{Br}_2$ (b) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$
- (c) $[\text{Cr}(\text{en})\text{Br}_4]^-$ (d) $[\text{Cr}(\text{en})\text{Br}_2]\text{Br}$
35. Which of the following complex species is not expected to exhibit optical isomerism ? (2013)
- (a) $[\text{Co}(\text{en})_3]^{3+}$ (b) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
- (c) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (d) $[\text{Co}(\text{en})(\text{NH}_3)\text{Cl}_2]^+$
36. The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1 , L_2 , L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is : (2014)
- (a) $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$ (b) $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$
- (c) $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$ (d) $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$
37. The number of geometric isomers that can exist for square planar $[\text{Pt}(\text{Cl})(\text{py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ is (py = pyridine) : (2015)
- (a) 4 (b) 6
- (c) 2 (d) 3
38. The pair having the same magnetic moment is: (2016)
- [At, No. : Cr = 24, Mn = 25, Fe = 26, Co = 27]
- (a) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
- (b) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
- (c) $[\text{CoCl}_4]^{2-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
- (d) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$
39. On treatment of 100 mL of 0.1 M solution of $\text{CoCl}_3 \cdot 6\text{H}_2\text{O}$ with excess of AgNO_3 ; 1.2×10^{22} ions are precipitated. The complex is (2017)
- (a) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
- (b) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
- (c) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
- (d) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
40. The oxidation states of Cr in $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Cr}(\text{C}_6\text{H}_6)_2]$ and $\text{K}_2[\text{Cr}(\text{CN})_2(\text{O})_2(\text{O}_2)(\text{NH}_3)]$ respectively are: (2018)
- (a) +3, 0 and +4 (b) +3, +4 and +6
- (c) +3, +2 and +4 (d) +3, 0 and +6

JEE MAINS ONLINE QUESTION

1. An octahedral complex of Co^{3+} is diamagnetic. The hybridization involved in the formation of the complex is:

Online 2014 SET (1)

- (a) sp^3d^2 (b) dsp^2
(c) d^2sp^3 (d) dsp^3d

2. The correct statement about the magnetic properties of $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ is : ($Z = 26$).

Online 2014 SET (1)

- (a) both are paramagnetic.
(b) both are diamagnetic.
(c) $[\text{Fe}(\text{CN})_6]^{3-}$ is diamagnetic, $[\text{FeF}_6]^{3-}$ is paramagnetic.
(d) $[\text{Fe}(\text{CN})_6]^{3-}$ is paramagnetic, $[\text{FeF}_6]^{3-}$ is diamagnetic.

3. Consider the coordination compound, $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$. In the formation of this complex, the species which acts as the Lewis acid is:

Online 2014 SET (2)

- (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) Co^{3+}
(c) Cl^- (d) NH_3

4. Which of the following name formula combinations is not correct?

Online 2014 SET (2)

Formula	Name
(a) $\text{K}[\text{Cr}(\text{NH}_3)_2\text{Cl}_4]$	Potassium diammine tetrachlorochromate (III)
(b) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})]\text{SO}_4$	Tetraammine aquaiodo cobalt (III) sulphate
(c) $\text{K}_2[\text{Pt}(\text{CN})_4]$	Potassium tetracyanoplatinate (II)
(d) $[\text{Mn}(\text{CN})_5]^{2-}$	Pentacyanomagnate (II) ion

5. Which one of the following complexes will most likely absorb visible light?

Online 2014 SET (3)

(At nos: Sc = 21, Ti = 22, V = 23, Zn = 30)

- (a) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Zn}(\text{NH}_3)_6]^{2+}$
(c) $[\text{Ti}(\text{NH}_3)_6]^{4+}$ (d) $[\text{V}(\text{NH}_3)_6]^{3+}$

6. Among the following species the one which causes the highest CFSE, Δ_o as a ligand is: Online 2014 SET (3)

- (a) F^- (b) NH_3
(c) CO (d) CN^-

7. An octahedral complex with molecular composition $\text{M.5NH}_3.\text{Cl.SO}_4$ has two isomers, A and B. The solution of A gives a white precipitate with AgNO_3 solution and the solution of B gives white precipitate with BaCl_2 solution. The type of isomerism exhibited by the complex is :

Online 2014 SET (4)

- (a) Linkage isomerism
(b) Coordinate isomerism
(c) Geometrical isomerism
(d) Ionisation isomerism

8. Among the following, identify the specific with an atom in +6 oxidation state :

Online 2014 SET (4)

- (a) CrO_2Cl_2 (b) $[\text{MnO}_4]^-$
(c) Cr_2O_3 (d) $[\text{Cr}(\text{CN})_6]^{3-}$

9. Nickel ($Z = 28$) combines with a uninegative monodentate ligand to form a diamagnetic complex $[\text{NiL}_4]^{2-}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively :

Online 2014 SET (4)

- (a) dsp^2 , one (b) dsp^2 , zero
(c) sp^3 , zero (d) sp^3 , two

10. When concentrated HCl is added to an aqueous solution of CoCl_2 , its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction ?

Online 2015 SET (1)

- (a) $[\text{CoCl}_4]^{2-}$ (b) $[\text{CoCl}_6]^{3-}$
(c) $[\text{CoCl}_6]^{4-}$ (d) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$

11. Which of the following is an example of homoleptic complex ?

Online 2016 SET (2)

- (a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (b) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
(c) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (d) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

12. $[\text{Co}_2(\text{CO})_8]$ displays : **Online 2017 SET (2)**
- (a) one Co – Co bond, six terminal CO and two bridging CO
 (b) one Co – Co bond, four terminal CO and four bridging CO
 (c) no Co – Co bond, six terminal CO and two bridging CO
 (d) no Co – Co bond, four terminal CO and four bridging CO
13. The correct combination is : **Online 2018 SET (1)**
- (a) $[\text{Ni}(\text{CN})_4]^{2-}$ - tetrahedral ; $[\text{Ni}(\text{CO})_4]$ - paramagnetic
 (b) $[\text{NiCl}_4]^{2-}$ - paramagnetic ; $[\text{Ni}(\text{CO})_4]$ - tetrahedral
 (c) $[\text{NiCl}_4]^{2-}$ - square-planar ; $[\text{Ni}(\text{CN})_4]^{2-}$ - paramagnetic
 (d) $[\text{NiCl}_4]^{2-}$ - diamagnetic ; $[\text{Ni}(\text{CO})_4]$ - square-planar
14. The correct order of spin-only magnetic moments among the following is : **Online 2018 SET (2)**
- (Atomic number : Mn =25, Co =27, Ni =28, Zn =30)
- (a) $[\text{ZnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 (b) $[\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (c) $[\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (d) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
15. The total number of possible isomers for square-planar $[\text{Pt}(\text{Cl})(\text{NO}_2)(\text{NO}_3)(\text{SCN})]^{2-}$ is : **Online 2018 SET (2)**
- (a) 8 (b) 12
 (c) 16 (d) 24
16. Which of the following complexes will show geometrical isomerism ? **Online 2018 SET (3)**
- (a) Aquachlorobis (ethylenediamine) cobalt (II) chloride
 (b) pentaquachlorochromium (III) chloride
 (c) potassium amminetrichloroplatinate (II)
 (d) potassium tris(oxalato)chromate (III)

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

- All questions marked “S” are single choice questions
 - All questions marked “M” are multiple choice questions
 - All questions marked “C” are comprehension based questions
 - All questions marked “A” are assertion–reason type questions
 - If both assertion and reason are correct and reason is the correct explanation of assertion.
 - If both assertion and reason are true but reason is not the correct explanation of assertion.
 - If assertion is true but reason is false.
 - If reason is true but assertion is false.
 - All questions marked “X” are matrix–match type questions
 - All questions marked “I” are integer type questions
-
- (S) The IUPAC name for $[\text{Co}(\text{ONO})(\text{NH}_3)_5]\text{SO}_4$ is
 - pentaammine (nitrito-O) cobalt (III) sulphate
 - pentaammine (nitroso-N) cobalt (III) sulphate
 - pentaammine (nitro-O) cobalt (III) sulphate
 - pentaammine (nitroyl-O) cobalt (III) sulphate
 - (S) The IUPAC name for $[\text{Co}(\text{NCS})(\text{NH}_3)_5]\text{Cl}_2$ is
 - pentaammine (isothiocyanato- η^2 N, S) cobalt (III) chloride
 - pentaammine (mercapto-N) cobalt (III) chloride
 - pentaammine (thiocyanato-N) cobalt (III) chloride
 - pentaamine (thiocyanato-S) cobalt (III) chloride
 - (S) The IUPAC name for $[(\text{NH}_3)_5\text{Cr}-(\text{OH})-\text{Cr}(\text{NH}_3)_5]^{5+}$ is
 - μ -hydroxo-bis (pentaammine dichromium) (5+) ion
 - μ -hydroxo-bis (decaamminedichromium) (5+) ion
 - μ -hydroxo-bis (octaamminechromium) (5+) ion
 - μ -hydroxo-bis [(pentaamminechromium) (III)]
 - (S) The IUPAC name for $[(\text{CO})_3\text{Fe}(\text{CO})_3\text{Fe}(\text{CO})_3]$ is
 - tri- μ -hexacarbonyl iron (II)
 - tri- μ -carbonyl-bis[tricarbonyliron (0)]
 - tri- μ -carbonyl-bis[tricarbonyldiiron (0)]
 - tri- μ -nanocarbonyl iron (III)
 - (S) Which of the following complex ions is octahedral ?
 - $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$
 - $[\text{Fe}(\text{NO}_3)_4]^-$
 - $[\text{Fe}(\text{CO})_2(\text{NO})_2]$
 - $[\text{Fe}(\text{NSiMe}_3)_2]_3$
 - (S) The IUPAC name for $[\text{Be}_4\text{O}(\text{CH}_3\text{COO})_6]$ is
 - basic beryllium acetate (II)
 - hexa- μ -acetato (O, O')- μ_4 -oxo-tetraberyllium (II)
 - hexa- μ -hexakis (acetato) beryllium (II)
 - hexaacetato- μ -oxo-beryllate (II)
 - (S) The IUPAC name for $[(\text{CO})_5\text{Mn}-\text{Mn}(\text{CO})_5]$ is
 - bis (pentacarbonyldimanganese)
 - bis [pentacarbonylmanganate (VIII)]
 - decacarbonyldimanganate (VIII)
 - bis [pentacarbonylmanganese (0)]
 - (S) The IUPAC name for $[\text{Pt}(\text{py})_4][\text{PtCl}_4]$ is
 - tetrakis(pyridine) platinum(II) tetrachloroplatinate (II)
 - tetrapyridine tetrachlorodiplatinum (IV)
 - tetrachloro tetrapyridinediplatinum (II)
 - tetrakis (pyridine) platinum (IV) tetrachloroplatinum (IV)
 - (S) The IUPAC name for $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ is
 - hexaamminecobalt (III) hexacyanochromium (VI)
 - hexaamminecobalt (III) hexacyanochromium (III)
 - hexacyanochromium (III) hexaamminecobalt (III)
 - hexacyanochromium cobalt hexaamminecobalt (VI)
 - (S) The formula of the complex potassium trichloro (ethylene) platinate (II) is
 - $\text{K}[\text{PtCl}_3(\text{C}_2\text{H}_4)]$
 - $\text{K}_2[\text{PtCl}_3(\text{C}_2\text{H}_4)_3]$
 - $\text{K}_4[\text{PtCl}_3(\text{C}_2\text{H}_4)]$
 - $\text{K}_3[\text{Pt}_2\text{Cl}_3(\text{C}_2\text{H}_4)_3]$

- 11. (S)** The formula of the complex triammine trinitrocobalt (III) is
(a) $[\text{Co}(\text{ONO})_3(\text{NH}_3)_3]$ (b) $[\text{Co}(\text{ONO}_2)_3(\text{NH}_3)_3]$
(c) $[\text{Co}(\text{ONO})_3(\text{NH}_3)_3]$ (d) $[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$
- 12. (S)** The formula of the complex sodium hydridotrimethoxoborate (III) is
(a) $\text{Na}_4[\text{B}_4\text{H}_2(\text{OCH}_3)_3]$ (b) $\text{Na}_2[\text{BH}(\text{OCH}_3)_3]$
(c) $\text{Na}[\text{BH}_2(\text{OCH}_3)_3]$ (d) $\text{Na}[\text{BH}(\text{OCH}_3)_3]$
- 13. (S)** The value of magnetic moment for a complex ion is 1.73 BM. The complex ion is
(a) $[\text{Co}(\text{NH}_3)_6]^{2+}$ (b) $[\text{MnF}_6]^{3-}$
(c) $[\text{Fe}(\text{CN})_5]^{2-}$ (d) $[\text{Mn}(\text{CN})_6]^{4-}$
- 14. (S)** The magnetic moment of a complex ion is 2.83 BM. The complex ion is
(a) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Cu}(\text{CN})_4]^{2-}$
(c) $[\text{V}(\text{H}_2\text{O})_6]^{3+}$ (d) $[\text{MnCl}_4]^{2-}$
- 15. (S)** Which of the following is a low-spin (spin-paired) complex?
(a) $[\text{Co}(\text{NH}_3)_6]^{2+}$ (b) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$
(c) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (d) $[\text{Co}(\text{NH}_3)_6]^{3+}$
- 16. (S)** Which of the following is a high-spin (spin-free) complex?
(a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Fe}(\text{CN})_6]^{4-}$
(c) $[\text{CoF}_6]^{3-}$ (d) $[\text{Zn}(\text{NH}_3)_6]^{2+}$
- 17. (S)** Which of the following complexes is most easily oxidized?
(a) $\text{Ni}(\text{CO})_4$ (b) $[\text{Mn}(\text{CN})_6]^{6-}$
(c) $\text{Cr}(\text{CO})_4$ (d) $\text{Fe}(\text{CO})_5$
- 18. (S)** Which of the following complexes is the most stable?
(a) $[\text{Ni}(\text{CN})_4]^{4-}$ (b) $[\text{Pd}(\text{CN})_4]^{4-}$
(c) $[\text{Ni}(\text{CN})_4]^{3-}$ (d) $\text{Fe}(\text{CO})_5$
- 19. (S)** Which of the following aquated metal ions has the highest paramagnetism?
(a) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
(c) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ (d) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
- 20. (S)** Which of the following species is expected to be colourless?
(a) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Ti}(\text{NO}_3)_4]$
(c) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (d) $[\text{Fe}(\text{CN})_6]^{4-}$
- 21. (S)** Tollens reagent contains
(a) AgNO_3 (b) AgOH
(c) $[\text{Ag}(\text{NH}_3)_2]^+$ (d) $[\text{Ag}(\text{NO}_3)_2]^+$
- 22. (S)** The chloride ion is detected by the formation of a white precipitate of AgCl on addition of aqueous AgNO_3 solution and by the dissolution of the precipitate on addition of excess ammonia solution. AgCl dissolves in aqueous ammonia due to the formation of water soluble
(a) $[\text{Ag}(\text{NH}_4)_2]^+$ (b) $[\text{Ag}(\text{NH}_3)_2]^+$
(c) $[\text{Ag}(\text{NH}_2)_2]^-$ (d) $[\text{Ag}(\text{OH})_2]^-$
- Comprehension**
- A complex compound of chromium contains five NH_3 molecules, one nitro group and two chloride ions for one Cr^{3+} cation. One molecule of this compound produces three ions in aq. solution, on reacting with excess of AgNO_3 solution, two moles of AgCl get precipitated.
- 23. (C)** The formula of the complex compound is :
(a) $[\text{CrCl}(\text{NO}_2)(\text{NH}_3)_4]\text{NH}_3\text{Cl}$
(b) $[\text{CrCl}(\text{NH}_3)_5]\text{Cl}.\text{NO}_2$
(c) $[\text{Cr}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$
(d) $[\text{Cr}(\text{NH}_3)_5]\text{NO}_2.\text{Cl}_2$
- 24. (C)** The types of isomerism shown by the complex compound is :
(a) geometrical, ionization
(b) ionization, linkage
(c) linkage, optical
(d) geometrical, optical
- 25. (C)** Magnetic moment of complex compound is :
(a) 0 BM (b) $\sqrt{24}$ BM
(c) $\sqrt{15}$ BM (d) $\sqrt{3}$ BM

Comprehension

An isomer of the complex $\text{Co(en)}_2(\text{H}_2\text{O})\text{ICl}_2$, on reaction with concentrated H_2SO_4 it suffers loss in weight and on reaction with AgNO_3 solution gives a yellow precipitate, which is insoluble in NH_3 solution.

26. (C) If all the ligands in the co-ordination sphere of the above complex are replaced by CN^- ion, then the magnetic moment of the complex ion will be :
 (a) 0.0 BM (b) 5.9 BM
 (c) 4.9 BM (d) 1.73 BM
27. (C) If one of mole of original complex is treated with excess $\text{Pb}(\text{NO}_3)_2$ solution, then the number of moles of white precipitate formed will be :
 (a) 2.0 (b) 1.0
 (c) 0.0 (d) 3.0
28. (C) Total number of space isomers of the formula of the above complex is :
 (a) 2 (b) 3
 (c) 4 (d) 1

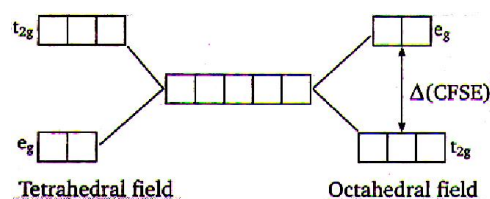
Comprehension

On the basis of stability of complex ion in the solution, complexes may be of two types perfect and imperfect complexes. The stability depends upon the extent of dissociation which in turn depends upon the strength of metal- ligand bond. The stability of complex also depends upon charge on central metal atom, basic nature of ligand, chelation, and natural metal ion and ligand according to HSAB principle ?

29. (C) Which complex is most stable ?
 (a) $[\text{Cu}(\text{CN})_2]^- K_d = 1 \times 10^{-16}$
 (b) $[\text{Fe}(\text{CN})_6]^{4-} K_d = 1 \times 10^{-37}$
 (c) $[\text{Fe}(\text{CN})_6]^{3-} K_d = 1 \times 10^{-44}$
 (d) $[\text{Ag}(\text{CN})_2]^- K_d = 1 \times 10^{-20}$

Comprehension

When degenerate d-orbitals of an isolated atom/ion come under influence of magnetic field of ligands, the degeneracy is lost. The two set t_{2g} (d_{xy} , d_{yz} , d_{xz}) and e_g (d_{z^2} , $d_{x^2-y^2}$) are either stabilized or destabilized depending upon the nature of magnetic field. It can be expressed diagrammatically as :



Value of CFSE depends upon nature of ligand and a spectrochemical series has been made experimentally, for tetrahedral complexes, Δ is about $4/9$ times to Δ_0 (CFSE for octahedral complex). This energy lies in visible region and i.e., why electronic transition are responsible for colour. Such transitions are not possible with d_0 and d^{10} configuration.

30. (C) The value of CFSE (Δ_0) for complexes given below follow the order :
 (I) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (II) $[\text{Rh}(\text{NH}_3)_6]^{3+}$
 (III) $[\text{Ir}(\text{NH}_3)_6]^{3+}$
 (a) $\text{I} < \text{II} < \text{III}$ (b) $\text{I} > \text{II} > \text{III}$
 (c) $\text{I} < \text{II} > \text{III}$ (d) $\text{I} = \text{II} = \text{III}$
31. (C) Cr^{3+} form four complexes with four different ligands which are $[\text{Cr}(\text{Cl})_6]^{3-}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and $[\text{Cr}(\text{CN})_6]^{3-}$. The order of CFSE (Δ_0) in these complexes is in the order :
 (a) $[\text{CrCl}_6]^{3-} = [\text{Cr}(\text{H}_2\text{O})_6]^{3+} = [\text{Cr}(\text{NH}_3)_6]^{3+} = [\text{Cr}(\text{CN})_6]^{3-}$
 (b) $[\text{CrCl}_6]^{3-} < [\text{Cr}(\text{H}_2\text{O})_6]^{3+} < [\text{Cr}(\text{NH}_3)_6]^{3+} < [\text{Cr}(\text{CN})_6]^{3-}$
 (c) $[\text{CrCl}_6]^{3-} > [\text{Cr}(\text{H}_2\text{O})_6]^{3+} > [\text{Cr}(\text{NH}_3)_6]^{3+} > [\text{Cr}(\text{CN})_6]^{3-}$
 (d) $[\text{CrCl}_6]^{3-} < [\text{Cr}(\text{H}_2\text{O})_6]^{3+} = [\text{Cr}(\text{NH}_3)_6]^{3+} < [\text{Cr}(\text{CN})_6]^{3-}$
32. (C) The d-orbitals, which are stabilized in an octahedral magnetic field, are :
 (a) d_{xy} and d_{z^2} (b) $d_{x^2-y^2}$ and d_{z^2}
 (c) d_{xy} , d_{xz} and d_{yz} (d) d_{z^2} only
33. (C) For an octahedral complex, which of the following d-electron configuration will give maximum CFSE ?
 (a) High spin $d^6 - 0.4 \Delta_0$ (b) Low spin $d^4 - 1.6 \Delta_0$
 (c) Low spin $d^5 - 2.0 \Delta_0$ (d) High spin $d^7 - 0.8 \Delta_0$

34. (C) $\text{Ti}^{3+}(\text{aq})$ is purple while $\text{Ti}^{4+}(\text{aq})$ is colourless because :
- There is no crystal field effect in Ti^{4+}
 - The energy difference between t_{2g} and e_g Ti^{4+} is quite high and does not fall in the visible region
 - Ti^{4+} has d^0 configuration
 - Ti^{4+} is very small in comparison to Ti^{3+} and hence does not absorb any radiation.
35. (C) Which of the following is correct arrangement of ligands in terms of the Dq values of their complexes with any particular 'hard' metal ion :
- $\text{Cl}^- < \text{F}^- < \text{NCS}^- < \text{NH}_3 < \text{CN}^-$
 - $\text{NH}_3 < \text{F}^- < \text{Cl}^- < \text{NCS}^- < \text{CN}^-$
 - $\text{Cl}^- < \text{F}^- < \text{NCS}^- < \text{CN}^- < \text{NH}_3$
 - $\text{NH}_3 < \text{CN}^- < \text{NCS}^- < \text{Cl}^- < \text{F}^-$
36. (C) The extent of crystal field splitting in octahedral complexes of the given metal with particular weak field ligand are :
- $\text{Fe(III)} < \text{Cr(III)} < \text{Rh(III)} < \text{Ir(III)}$
 - $\text{Cr(III)} < \text{Fe(III)} < \text{Rh(III)} < \text{Ir(III)}$
 - $\text{Ir(III)} < \text{Rh(III)} < \text{Fe(III)} < \text{Cr(III)}$
 - $\text{Fe(III)} < \text{Cr(III)} < \text{Rh(III)} < \text{Ir(III)}$
37. (M) Complex ions $[\text{NiCl}_6]^{4-}$, $[\text{Ni}(\text{CN})_6]^{4-}$ similar in their given properties :
- oxidation state, geometry
 - co-ordination number, EAN
 - magnetic moment, geometry
 - Stability, Colour
38. (M) Select correct statement(s) regarding given complex :
- $[\text{Fe}(\text{CO})_5]$, the orbitals used for hybridization in Fe atom are $s, p_x, p_y, p_z, d_{z^2}$ and it is high spin complex.
 - $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, the orbitals used for hybridization on Pt atom are $s, p_x, p_y, d_{x^2-y^2}$, and it is low spin complex.
 - $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, the orbitals used for hybridization in Cr atom are $s, p_x, p_y, p_z, d_{x^2-y^2}$, and it is high spin complex.
 - $\text{Ni}(\text{CO})_4$, the orbitals used for hybridization in Ni atom are, s, p_x, p_y, p_z , and it is low spin complex.
39. (M) Complex compound $[\text{Co}(\text{SCN})_2(\text{NH}_3)_4]\text{Cl}$ exhibits :
- ionization isomerism
 - geometrical isomerism
 - optical isomerism
 - linkage isomerism
40. (M) Which of the following statement(s) is/are correct ?
- The oxidation state of iron in sodium nitro prusside $\text{Na}_2[\text{Fe}(\text{CN})_5(\text{NO})]$ is + II
 - $[\text{Ag}(\text{NH}_3)_2]^+$ is linear in shape
 - In $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, Fe is d^2sp^3 hybridized
 - In $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ complex ion, Co is d^2sp^3 hybridized
41. (M) Which one of the following statement(s) is/are false ?
- Weak ligands like F^- , Cl^- , and OH^- usually form low spin complexes
 - Strong ligands like CN^- and NO_2^- , generally form high spin complexes
 - $[\text{FeF}_6]^{3-}$ is high spin complex
 - $[\text{Ni}(\text{CO})_4]$ is high spin complex.
42. (M) For which of the following d^n configuration of octahedral complex(es), cannot exist in both high spin and low spin forms ?
- d^3
 - d^5
 - d^6
 - d^8
43. (M) Which of the following is correct about Tetraamminedithiocyanato-s cobalt (III) tris(oxalato) cobaltage (III) ?
- Formula of the complex is $[\text{Co}(\text{SCN})_2(\text{NH}_3)_4][\text{Co}(\text{ox})_3]$
 - It is chelating complex and show linkage isomerism
 - It shows optical isomerism
 - It shows geometrical isomerism
44. (M) Select the correct statement :
- Chelation effect is maximum for five and six membered rings.
 - Greater the charge on the central metal cation, greater the value of Δ (CFSE)
 - In complex ion $[\text{CoF}_6]^{3-}$, F^- is a weak field ligand, so that $\Delta_{\text{oct}} < P$ (Pairing energy) and it is low spin complex.
 - $[\text{CoCl}_2(\text{NH}_3)_2(\text{en})]^{\oplus}$ complex ion will have four different isomers.

45. (M) In test of NO_3^- ion, the dark brown ring complex is formed, which is true of this complex ?
- (a) The colour is due to charge transfer spectra
- (b) Iron and NO both have +1 charge
- (c) The complex species can be represented as $[\text{Fe}^{\text{I}}(\text{H}_2\text{O})_5\text{NO}]^{2+}$
- (d) Iron has +2 oxidation state and NO is neutral.

46. (M) The complex(es) which is/are blue in colour :

- (a) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (b) $\text{Zn}_2[\text{Fe}(\text{CN})_6]$
- (c) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$ (d) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$

Column -I and Column-II contains four entries each. Entries of Column-I are to be matched with one or more than one entries of column II. Each entry of Column-I may have the matching with one or more than one entries of Column-II

47. (X)
- | Column-I | Column-II |
|--|----------------------------|
| (A) $[\text{Cr}(\text{CN})_3(\text{NO}_2)_3]^{4-}$ | (P) Outer orbital complex |
| (B) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ | (Q) Inner orbital complex |
| (C) $[\text{Fe}(\text{EDTA})]^-$ | (R) $\mu = 2.83$ BM |
| (D) $[\text{Ni}(\text{en})_3](\text{NO}_3)_2$ | (S) Shows optical activity |
| | (T) $\mu = 0$ |

48. (X)
- | Column-I | Column-II |
|--|----------------------------|
| (A) $[\text{Fe}(\text{CN})_6]^{4-}$ | (P) Paramagnetic |
| (B) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ | (Q) Diamagnetic |
| (C) $[\text{Cu}(\text{NH}_3)_6]^{2+}$ | (R) Inner orbital complex |
| (D) $[\text{Ni}(\text{CN})_6]^{4-}$ | (S) Outer orbital complex. |

49. (X)
- | Column-I | Column-II |
|--|--|
| (Molar conductance at infinite dilution) | (Complex compound) |
| (A) 229 | (P) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$ |
| (B) 0 | (Q) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ |
| (C) 404 | (R) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$ |
| (D) 523 | (S) $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$ |

50. (X)
- | Column-I | Column-II |
|-------------------------------------|--------------------------------|
| (Complex ions) | (Number of unpaired electrons) |
| (A) $[\text{CrF}_6]^{4-}$ | (P) One |
| (B) $[\text{MnF}_6]^{4-}$ | (Q) Two |
| (C) $[\text{Cr}(\text{CN})_6]^{4-}$ | (R) Three |
| (D) $[\text{Mn}(\text{CN})_6]^{4-}$ | (S) Four |
| | (T) Five |

51. (X)
- | Column-I | Column-II |
|-------------------------------------|---------------------------------|
| (Complexes) | (Hybridization of central atom) |
| (A) $\text{Ni}(\text{CO})_4$ | (P) sp^3 |
| (B) $[\text{Ni}(\text{CN})_4]^{2-}$ | (Q) dsp^2 |
| (C) $[\text{Fe}(\text{CN})_6]^{4-}$ | (R) sp^3d^2 |
| (D) $[\text{MnF}_6]^{4-}$ | (S) d^2sp^3 |

52. (A) **Assertion :** $[\text{Ti}(\text{H}_2\text{O})_6]^{4+}$ is coloured while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless.

Reason : d-d transition is not possible in $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$.

- (a) A (b) B
- (c) C (d) D

53. (A) **Assertion :** $[\text{Fe}(\text{EDTA})]^-$ complex is octahedral in shape.

Reason : EDTA is a hexadentate ligand and undergoing sp^3d^2 hybridisation.

- (a) A (b) B
- (c) C (d) D

54. (A) **Assertion :** Oxidation state Fe in $\text{Fe}(\text{CO})_5$ is zero.

Reason : Synergic bonding takes place in this metal carbonyl complex.

- (a) A (b) B
- (c) C (d) D

55. (A) **Assertion :** Zeise's salt is a π -bonded organometallic compound.

Reason : Zeise's salt contain C_2H_4 molecules as one of the ligand.

- (a) A (b) B
- (c) C (d) D

- 56. (A) Assertion :** $[\text{CoCl}_3(\text{NH}_3)_3]$ does not give white precipitate with AgNO_3 solution.
Reason : $[\text{CoCl}_3(\text{NH}_3)_3]$ complex is optically inactive.
(a) A (b) B
(c) C (d) D
- 57. (A) Assertion :** Hydrazine is a neutral ligand.
Reason : It has two N as donor atoms and behaves as a chelating ligand.
(a) A (b) B
(c) C (d) D
- 58. (I)** What is CFSE of complex ion $[\text{FeF}_6]^{4-}$ in terms of Dq ?
- 59. (I)** How many electrons are present in t_{2g} set of d-orbitals of central metal cation in $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})] \text{SO}_4$ brown ring complex ?
- 60. (I)** Calculate value of “x ÷ y” if x is the total number of σ bonds and y is total number of π bonds in ligand EDTA and phenanthroline.
{EDTA = Ethylene diamine tetraacetate, phen = 1, 10 – N, N-Phenanthroline}
- 61. (I)** Total number of complexes among the following which are optically active ?
(i) $[\text{Cr}(\text{Ox})_3]^{3-}$
(ii) $\text{cis-}[\text{Pt}(\text{Cl}_2)(\text{en})]$
(iii) $\text{cis-}[\text{Rh}(\text{Cl}_2)(\text{NH}_3)_4]^+$
(iv) $[\text{Ru}(\text{dipy})_3]^{3+}$
(v) $\text{cis-}[\text{Co}(\text{NO}_2)_3(\text{dien})]$
(vi) $\text{Trans-}[\text{Co}(\text{NO}_2)_3(\text{dien})]$
(vii) $\text{cis-}[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$

EXERCISE - 4: PREVIOUS YEAR JEE ADVANCED QUESTION

- Match each set of hybrid orbitals from LIST-I with complex(es) given in LIST-II. (2018)

LIST-I	LIST-II
(P) dsp^2	(1) $[FeF_6]^{4-}$
(Q) sp^3	(2) $[Ti(H_2O)_3Cl_3]$
(R) sp^3d^2	(3) $[Cr(NH_3)_6]^{3+}$
(S) d^2sp^3	(4) $[FeCl_4]^{2-}$
	(5) $Ni(CO)_4$
	(6) $[Ni(CN)_4]^{2-}$

The correct option is

 - P - 5; Q - 4,6; R - 2,3; S - 1
 - P - 5,6; Q - 4; R - 3; S - 1,2
 - P - 6; Q - 4, 5; R - 1; S - 2,3
 - P - 4,6; Q - 5,6; R - 1,2; S - 3
- The geometries of the ammonia complexes of Ni^{2+} , Pt^{2+} and Zn^{2+} , respectively, are (2016)
 - octahedral, square planar and tetrahedral
 - square planar, octahedral and tetrahedral
 - tetrahedral, square planar and octahedral
 - octahedral, tetrahedral and square planar
- Among $[NiCO_4]$, $[NiCl_4]^{2-}$, $[Co(NH_3)_4Cl_2]Cl$, $Na_3[CoF_6]$, Na_2O_2 and CsO_2 , the total number of paramagnetic compounds is (2016)
 - 2
 - 3
 - 4
 - 5
- Consider the following complex ions, P, Q and R.

$$P = [FeF_6]^{3-}, Q = [V(H_2O)_6]^{2+} \text{ and } R = [Fe(H_2O)_6]^{2+}$$

The correct order of the complex ions, according to their spin-only magnetic moment values (in BM) is (2013)

 - $R < Q < P$
 - $Q < R < P$
 - $R < P < Q$
 - $Q < P < R$
- As per IUPAC nomenclature, the name of the complex $[Co(H_2O)_4(NH_3)_2]Cl_3$ is (2012)
 - tetraaquadiamminecobalt (III) chloride
 - tetraaquadiamminecobalt (III) chloride
 - diaminetetraaquacobalt (III) chloride
 - diamminetetraaquacobalt (III) chloride
- $NiCl_2 \cdot 2P(C_2H_5)_2(C_6H_5)_2$ exhibits temperature dependent magnetic behaviour (paramagnetic/diamagnetic) the coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively (2012)
 - tetrahedral and tetrahedral
 - square planar and square planar
 - tetrahedral and square planar
 - square planar and tetrahedral
- The spin only magnetic moment value (in Bohr magneton units) of $Cr(CO)_6$ is (2009)
 - 0
 - 2.84
 - 4.90
 - 5.92
- Among the following, the coloured compound is (2008)
 - $CuCl$
 - $K_3[Cu(CN)_4]$
 - CuF_2
 - $[Cu(CH_3CN)_4]BF_4$
- Both $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ are diamagnetic. The hybridisations of nickel in these complexes, respectively, are (2008)
 - sp^3, sp^3
 - sp^3, dsp^2
 - dsp^2, sp^3
 - dsp^2, dsp^2
- The IUPAC name of $[Ni(NH_3)_4][NiCl_4]$ is (2008)
 - Tetrachloronickel (II)-tetraamminenickel (II)
 - Tetraamminenickel (II)-tetrachloronickel (II)
 - Tetraamminenickel (II)-tetrachloronickelate (II)
 - Tetrachloronickel (II)-tetraamminenickelate (0)
- Among the following metal carbonyls, the C—O bond order is lowest in (2007)
 - $[Mn(CO)_6]^+$
 - $[Fe(CO)_5]$
 - $[Cr(CO)_6]$
 - $[V(CO)_6]^-$
- If the bond length of CO bond in carbon monoxide is 1.128 Å, then what is the value of CO bond length in $Fe(CO)_5$?
 - 1.15 Å
 - 1.128 Å
 - 1.72 Å
 - 1.118 Å

13. Which kind of isomerism is exhibited by octahedral $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{Cl}$? (2005)
- (a) Geometrical and Ionization
(b) Geometrical and Optical
(c) Optical and Ionization
(d) Geometrical only
14. The species having tetrahedral shape is (2004)
- (a) $[\text{PdCl}_4]^{2-}$ (b) $[\text{Ni}(\text{CN})_4]^{2-}$
(c) $[\text{Pd}(\text{CN})_4]^{2-}$ (d) $[\text{NiCl}_4]^{2-}$
15. The compound having tetrahedral geometry is (2004)
- (a) $[\text{Ni}(\text{CN})_4]^{2-}$ (b) $[\text{Pd}(\text{CN})_4]^{2-}$
(c) $[\text{PdCl}_4]^{2-}$ (d) $[\text{NiCl}_4]^{2-}$
16. Spin only magnetic moment of the compound $\text{Hg}[\text{Co}(\text{SCN})_4]$ is (2004)
- (a) $\sqrt{3}$ (b) $\sqrt{15}$
(c) $\sqrt{24}$ (d) $\sqrt{8}$
17. Mixture X = 0.02 mole of $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and 0.02 mole of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ was prepared in 2L of solution.
1 L of mixture X + excess $\text{AgNO}_3 \rightarrow \text{Y}$
1 L of mixture X + excess $\text{BaCl}_2 \rightarrow \text{Z}$
Number of moles of Y and Z are (2003)
- (a) 0.01, 0.01 (b) 0.02, 0.01
(c) 0.01, 0.02 (d) 0.02, 0.02
18. The complex ion which has no d electrons in the central metal atom is (2001)
- (a) $[\text{MnO}_4]^-$ (b) $[\text{Co}(\text{NH}_3)_6]^{3+}$
(c) $[\text{Fe}(\text{CN})_6]^{3-}$ (d) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
19. The geometry of $\text{Ni}(\text{CO})_4$ and $\text{Ni}(\text{PPh}_3)_2\text{Cl}_2$ are (1999)
- (a) both square planar
(b) tetrahedral and square planar, respectively
(c) both tetrahedral
(d) square planar and tetrahedral, respectively
20. Which of the following is an organometallic compound? (1998)
- (a) Lithium methoxide (b) Lithium acetate
(c) Lithium dimethylamide (d) Methyl lithium
21. Among the following, the compound that is both paramagnetic and coloured is (1997)
- (a) $\text{K}_2\text{Cr}_2\text{O}_7$ (b) $(\text{NH}_4)_2[\text{TiCl}_6]$
(c) CoSO_4 (d) $\text{K}_3[\text{Cu}(\text{CN})_4]$
22. Which compound is formed when excess of KCN is added to aqueous solution of copper sulphate? (1996)
- (a) $\text{Cu}(\text{CN})_2$ (b) $\text{K}_2[\text{Cu}(\text{CN})_4]$
(c) $\text{K}[\text{Cu}(\text{CN})_2]$ (d) $\text{K}_3[\text{Cu}(\text{CN})_4]$
23. Among the following ions, which one has the highest paramagnetism? (1993)
- (a) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
(c) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (d) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
24. Amongst $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ (1991)
- (a) $[\text{Ni}(\text{CO})_4]$ and $[\text{NiCl}_4]^{2-}$ are diamagnetic $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic
(b) $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CO})_4]$ is paramagnetic
(c) $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic
(d) $[\text{Ni}(\text{CO})_4]$ is diamagnetic and $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are paramagnetic
25. Amongst the following, the lowest degree of paramagnetism per mole of the compound at 298 K will be shown by (1988)
- (a) $\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$ (b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
(c) $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$ (d) $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$
26. The pair (s) of coordination complexes/ions exhibiting the same kind of isomerism is (are) (2013)
- (a) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ and $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
(b) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ and $[\text{Pt}(\text{NH}_3)_2(\text{H}_2\text{O})\text{Cl}]^+$
(c) $[\text{CoBr}_2\text{Cl}_2]^{2-}$ and $[\text{PtBr}_2\text{Cl}_2]^{2-}$
(d) $[\text{Pt}(\text{NH}_3)_3(\text{NO}_3)]\text{Cl}$ and $[\text{Pt}(\text{NH}_3)_3\text{Cl}]\text{Br}$
27. The compound(s) that exhibit(s) geometrical isomerism is (are) (2009)
- (a) $[\text{Pt}(\text{en})\text{Cl}_2]$ (b) $[\text{Pt}(\text{en})_2]\text{Cl}_2$
(c) $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$ (d) $[\text{Pt}(\text{NH}_3)_2]\text{Cl}_2$

28. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $\text{MCl}_2 \cdot 6\text{H}_2\text{O}$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1:3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y.

Among the following options, which statement(s) is (are) correct ? (2017)

- (a) The hybridization of the central metal ion in Y is d^2sp^3
- (b) When X and Z are in equilibrium at 0°C , the colour of the solution is pink
- (c) Z is a tetrahedral complex
- (d) Addition of silver nitrate to Y gives only two equivalents of silver chloride.
29. The correct statement(s) regarding the binary transition metal carbonyl compounds is (are)
- (Atomic numbers : Fe = 26, Ni = 28) (2018)
- (a) Total number of valence shell electrons at metal centre in $\text{Fe}(\text{CO})_5$ or $\text{Ni}(\text{CO})_4$ is 16
- (b) These are predominantly low spin in nature.
- (c) Metal-carbon bond strengthens when the oxidation state of the metal is lowered
- (d) The carbonyl C-O bond weakens when the oxidation state of the metal is increased

Match the following

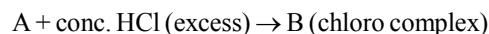
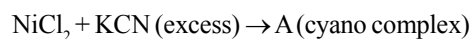
30. Match the complexes in column I with their properties listed in column II. (2007)

Column I	Column II
(A) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$	(p) geometrical isomers
(B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	(q) paramagnetic
(C) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}$	(r) diamagnetic
(D) $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$	(s) metal ion with +2 oxidation state

Comprehensions (Ques. 31-33)

Read the passage given below and answer the questions that follow (2006)

The coordination number of Ni^{2+} is 4.



31. The IUPAC name of A and B are
- (a) potassium tetracyanonickelate (II), potassiumtetrachloridonickelate (II)
- (b) potassiumtetracyanonickelate (II) tetrachloridopotassiumnickelate (II)
- (c) tetracyanonickel (II), tetrachloridonickel (II)
- (d) potassiumtetracyanonickel (II), potassiumtetrachloridonickel (II).
32. Predict the magnetic nature of A and B.
- (a) Both are diamagnetic
- (b) A is diamagnetic and B is paramagnetic with one unpaired electron
- (c) A is diamagnetic and B is paramagnetic with two unpaired electrons
- (d) Both are paramagnetic.
33. The hybridisation of A and B are
- (a) dsp^2 , sp^3 (b) sp^3 , sp^3
- (c) dsp^2 , dsp^2 (d) sp^3d^2 , d^2sp^3

Assertion & Reasons

- (A) If both ASSERTION and REASON are true and reason is the correct explanation of the assertion.
- (B) If both ASSERTION and REASON are true but reason is not the correct explanation of the assertion.
- (C) If ASSERTION is true but REASON is false.
- (D) If ASSERTION is false but REASON is true.
- (E) If both ASSERTION and REASON are false.

34. **Assertion** : $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ is paramagnetic
- Reason** : The Fe in $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ has three unpaired electrons (2008)
- (a) A (b) B (c) C
- (d) D (e) E

35. **Assertion** : The geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ are optically inactive.
Reason : Both geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ possess axis of symmetry. (2008)
- (a) A (b) B (c) C
(d) D (e) E

Fill in the Blanks

36. The type of magnetism exhibited by $[Mn(H_2O)_6]^{2+}$ ion is (1994)
37. The IUPAC name of $[Co(NH_3)_6]Cl_3$ is (1994)
- True/False**
38. Both potassium ferrocyanide and potassium ferricyanide are diamagnetic. (1989)
39. The electron density in the xy plane in $3d_{x^2-y^2}$ orbital is zero. (1986)

Subjective Question

40. Among the species given below, the total number of diamagnetic species is
H atom, NO_2 monomer, O_2^- (superoxide), dimeric sulphur in vapour phase, Mn_3O_4 , $[FeCl_4]^-$, $(NH_4)_2[NiCl_4]$, K_2MnO_4 , K_2CrO_4 (2018)
41. In dilute aqueous H_2SO_4 , the complex diaquodioxalatoferate(II) is oxidized by MnO_4^- . For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is (2015)
42. $EDTA^{4-}$ is ethylenediaminetetraacetate ion. The total number of N—Co—O bond angle in $[Co(EDTA)]^-$ complex ion is (2013)
43. $NiCl_2$ in the presence of dimethyl glyoxime (DMG) gives a complex which precipitates in the presence of NH_4OH , giving a bright red colour. (2004)
- (a) Draw its structure and show H-bonding
(b) Give oxidation state of Ni and its hybridisation
(c) Predict whether it is paramagnetic or diamagnetic.
44. Write the IUPAC name of the compound $K_2[Cr(NO)(CN)_4(NH_3)]$. Spin magnetic moment of the complex $\mu = 1.73$ BM. Give the structure of anion. (2003)
45. Deduce the structures of $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ considering the hybridisation of the metal ion. Calculate the magnetic moment (spin only) of the species. (2002)
46. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms A and B. The form A reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas B gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of A and B and state the hybridisation of chromium in each. Calculate their magnetic moments (spin-only value). (2001)
47. Draw the structures of $[Co(NH_3)_6]^{3+}$, $[Ni(CNH_3)_4]^{2-}$ and $[Ni(CO)_4]$. Write the hybridisation of atomic orbitals of the transition metal in each case. (2000)
48. A, B and C are three complexes of chromium (III) with the empirical formula $H_{12}O_6Cl_3Cr$. All the three complexes have water and chloride ion as ligands. Complex A does not react with concentrated H_2SO_4 , whereas complexes B and C lose 6.75% and 13.5% of their original mass, respectively, on treatment with concentrated H_2SO_4 . Identify A, B and C. (1999)
49. Write the formulae of the following complexes :
(i) Pentamminechlorocobalt (III) ion
(ii) Lithium tetrahydridoaluminate (III) (1997)
50. Write the IUPAC name for $[Cr(NH_3)_5CO_3]Cl$ (1996)
51. Write the IUPAC name of the following compounds
(i) $[Co(NH_3)_5ONO]Cl_2$ (ii) $K_3[Cr(CN)_6]$ (1995)
52. Identify the complexes which are expected to be coloured. Explain
(i) $[Ti(NO_3)_4]$ (ii) $[Cu(NCCH_3)]^+BF_4^-$
(iii) $[Cr(NH_3)_6]Cl_3$ (iv) $K_3[VF_6]$ (1994)
53. Give reasons in two or three sentences only for the following :
“The species $[CuCl_4]^{2-}$ exists, while $[CuI_4]^{2-}$ does not exist.” (1992)

ANSWER KEY

EXERCISE - 1 : (Basic Objective Questions)

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (d) | 3. (b) | 4. (d) | 5. (d) | 6. (c) | 7. (b) | 8. (c) | 9. (d) | 10. (b) |
| 11. (d) | 12. (c) | 13. (a) | 14. (c) | 15. (d) | 16. (b) | 17. (c) | 18. (d) | 19. (a) | 20. (c) |
| 21. (b) | 22. (b) | 23. (a) | 24. (c) | 25. (d) | 26. (c) | 27. (b) | 28. (d) | 29. (a) | 30. (c) |
| 31. (c) | 32. (d) | 33. (b) | 34. (c) | 35. (a) | 36. (d) | 37. (a) | 38. (c) | 39. (d) | 40. (a) |
| 41. (d) | 42. (b) | 43. (d) | 44. (c) | 45. (d) | 46. (b) | 47. (d) | 48. (c) | 49. (d) | 50. (d) |
| 51. (d) | 52. (b) | 53. (c) | 54. (d) | 55. (c) | 56. (a) | 57. (d) | 58. (c) | 59. (c) | 60. (d) |
| 61. (b) | 62. (d) | 63. (d) | 64. (c) | 65. (b) | 66. (d) | 67. (d) | 68. (a) | 69. (a) | 70. (a) |
| 71. (b) | | | | | | | | | |

Exercise - 2 : (Previous Year JEE Mains Questions)

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (d) | 3. (a) | 4. (c) | 5. (b) | 6. (c) | 7. (c) | 8. (b) | 9. (c) | 10. (b) |
| 11. (d) | 12. (b) | 13. (b) | 14. (d) | 15. (a) | 16. (d) | 17. (b) | 18. (c) | 19. (a) | 20. (d) |
| 21. (c) | 22. (d) | 23. (a) | 24. (b) | 25. (d) | 26. (d) | 27. (a) | 28. (c) | 29. (a) | 30. (d) |
| 31. (a) | 32. (c) | 33. (b) | 34. (a) | 35. (c) | 36. (d) | 37. (c) | 38. (a) | 39. (d) | 40. (d) |

JEE Mains Online

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|--------|--------|--------|---------|
| 1. (c) | 2. (a) | 3. (b) | 4. (d) | 5. (d) | 6. (c) | 7. (d) | 8. (a) | 9. (b) | 10. (a) |
| 11. (a) | 12. (a) | 13. (b) | 14. (d) | 15. (b) | 16. (a) | | | | |

Exercise - 3 : (Advanced Objective Questions)

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|--------------------------------------|-------------|-------------|------------|----------------------------------|-------------|-------------|-----------|---------|---------|
| 1. (a) | 2. (c) | 3. (d) | 4. (b) | 5. (a) | 6. (b) | 7. (d) | 8. (a) | 9. (b) | 10. (a) |
| 11. (c) | 12. (d) | 13. (d) | 14. (c) | 15. (d) | 16. (c) | 17. (b) | 18. (d) | 19. (b) | 20. (b) |
| 21. (c) | 22. (b) | 23. (c) | 24. (b) | 25. (c) | 26. (a) | 27. (c) | 28. (b) | 29. (c) | 30. (a) |
| 31. (b) | 32. (c) | 33. (c) | 34. (c) | 35. (a) | 36. (a) | 37. (a,b,c) | 38. (b,d) | | |
| 39. (a,b,d) | 40. (a,b,d) | 41. (a,b,d) | 42. (a,d) | 43. (b,c,d) | 44. (a,b,d) | 45. (a,b,c) | 46. (a,d) | | |
| 47. (A→Q,R; B→Q,S,T; C→Q,S; D→P,R,S) | | | | 48. (A→Q,R; B→P,S; C→P,S; D→P,S) | | | | | |
| 49. (A→R; B→Q; C→P; D→S) | | | | 50. (A→S; B→T; C→Q; D→P) | | | | | |
| 51. (A→P; B→Q; C→S; D→R) | 52. (d) | 53. (c) | 54. (b) | 55. (a) | 56. (b) | 57. (c) | | | |
| 58. (0004) | 59. (0005) | 60. (0005) | 61. (0002) | | | | | | |

Exercise - 4 : (Previous Year JEE Advanced Questions)

1. (c) 2. (a) 3. (b) 4. (b) 5. (d) 6. (c) 7. (a) 8. (c) 9. (b) 10. (c)
11. (d) 12. (a) 13. (a) 14. (d) 15. (d) 16. (b) 17. (a) 18. (a) 19. (c) 20. (d)
21. (c) 22. (d) 23. (b) 24. (c) 25. (b) 26. (b,d) 27. (c,d) 28. (a,b,c) 29. (b,c)
30. $A \rightarrow p, q, s$; $B \rightarrow p, r, s$; $C \rightarrow q, s$; $D \rightarrow q, s$ 31. (a) 32. (c) 33. (a) 34. (a) 35. (b)
36. paramagnetism 37. hexaammine cobalt (III) chloride 38. (F) 39. (F) 40. (0001)
41. (0008) 42. (0008) 45. $\mu = 0$ 46. $\mu = 0$
48. $A = [\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $B = [\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$, $C = [\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
49. (i) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ (ii) $\text{Li}[\text{AlH}_4]$ 50. pentaaminecarbonatochromium (III) chloride.
51. (i) pentaamminenitritocobalt (III) chloride, (ii) potassium hexacyanochromate (III) 52. (iii) & (iv)

Dream on !!

